

A vision–language pretrained transformer for versatile clinical respiratory disease applications

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General artificial intelligence models have unique challenges in clinical practice when applied to diverse modalities and complex clinical tasks. Here we present MedMPT, a versatile, clinically oriented pretrained model tailored for respiratory healthcare, trained on 154,274 pairs of chest computed-tomography scans and radiograph reports. MedMPT adopts self-supervised learning to acquire medical insights and is capable of handling multimodal clinical data and supporting various clinical tasks aligned with clinical workflows. We evaluate the performance of MedMPT on a broad spectrum of chest-related pathological conditions, involving common medical modalities such as computed tomography images, radiology reports, laboratory tests and drug relationship graphs. MedMPT consistently outperforms the state-of-the-art multimodal pretrained models in the medical domain, achieving significant improvements in diverse clinical tasks. Extensive analysis indicates that MedMPT effectively harnesses the potential of medical data, showing both data and parameter efficiency and offering explainable insights for decision-making. MedMPT highlights the potential of multimodal pretrained models in the realm of general-purpose artificial intelligence for clinical practice.

Lung diseases, including a wide range of pathological conditions affecting the chest—such as chronic obstructive pulmonary disease, lower respiratory tract infections and lung cancer—are a leading cause of morbidity and mortality worldwide¹. Timely diagnosis and management are crucial in improving health service performance and patient outcomes. Previous work suggests that artificial intelligence (AI) has the potential to advance lung disease diagnosis and management, for example, by enabling lung cancer screening based on chest computed-tomography

(CT) images² and generating radiology reports for chest X-rays³, thereby improving the effectiveness and efficiency of healthcare. From a clinical perspective, the diagnosis and management of lung diseases present a range of complexities due to the diverse nature of these conditions and the intricate healthcare systems involved, requiring a combination of various factors (such as medical history, physical examinations, imaging and laboratory tests) and necessitating a multidisciplinary approach, involving pulmonologists, radiologists and oncologists.

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However, there has been limited AI work that can comprehensively analyse multiple modalities, accurately provide medical insight for decision-making, and widely support diverse tasks in the clinical workflow within a unified framework, resulting in poor reliability and challenges for clinical translation of existing models in practice. Therefore, navigating these limitations to develop a universal model aligned with the clinical workflow for lung diseases is essential to improve patient treatment outcomes and enhance the efficiency of healthcare systems.

Recent advances in large pretrained models, such as ChatGPT⁴ and SAM⁵, have shown impressive generalizability and versatility and underscored the importance of general-purpose AI models. By learning general knowledge from massive data, these pretrained models can be effectively and efficiently adapted to various downstream tasks, without collecting and labelling a large amount of data for each specific task. These advantages of pretrained AI models are attractive for the medical field⁶, offering a solution to mitigate the challenges associated with extensive annotation requirements and to meet the demand for enhanced generalizability^{7–13} in healthcare.

However, the landscape of real-world clinical workflows is more challenging than anticipated, as the data modalities involved and the kinds of tasks in clinical practice are much more diverse, leaving the potential of existing pretrained AI models largely untapped in medicine. Most existing medical pretrained models are adaptations of general-domain pretrained models that have been fine-tuned with medical data. These models primarily focus on the tasks involving either image data^{9–11} or text data^{8,12,14}. In contrast, medical data inherently encompass multiple modalities¹⁵, such as CT images, medical texts, electronic records, laboratory tests, genomics and drug–drug relationships. In clinical practice, experts synthesize information from various modalities to comprehensively assess the patient's clinical condition and perform complex clinical tasks. Therefore, incorporating a variety of heterogeneous modalities is essential for comprehensive and thorough healthcare analysis. Another gap between existing medical models and real-world clinical workflow lies in the types of downstream tasks they support. Existing pretrained models primarily focus on the tasks within an isolated part of the clinical workflow, such as universal lesion segmentation^{9,10}, multiple diseases diagnosis¹¹ or answering generalized medical questions⁸. However, in the real-world clinical workflow, the tasks involved are much more diverse, continuous and coherent, covering various stages from a patient's arrival at the hospital to their discharge. For example, in the clinical workflow for patients with suspected lung cancer, clinicians usually order CT imaging and biochemical examinations, with radiologists interpreting the CT images and providing reports. This enables clinicians to comprehensively assess the patient's condition based on the examination results and offer a preliminary diagnosis and medical advice, potentially including medication prescription or recommendations for further procedures, such as needle biopsy or surgery¹⁶. Therefore, we argue that most current work mainly concentrates on data- and technique-driven validation tasks, failing to cover the entire clinical workflow and falling short of real clinical demand. The limitations in the modalities and tasks supported by existing medical pretrained models highlight the need to develop more effective, practical and clinically aligned AI models capable of analysing diverse medical data modalities and supporting a wide range of clinical tasks¹⁵.

Here we present a multimodal pretrained transformer (MedMPT) tailored to chest-related medical data using 154,274 pairs of chest CT scans and their corresponding reports through self-supervised learning. We evaluate its effectiveness, generalizability and explainability in diverse clinical tasks with multimodal data, such as CT images, free-text reports, demographic information, laboratory tests and drug relationship graphs. Self-supervised learning is an approach that leverages the inherent structure or patterns within the data, using the data itself as supervision to autonomously learn meaningful

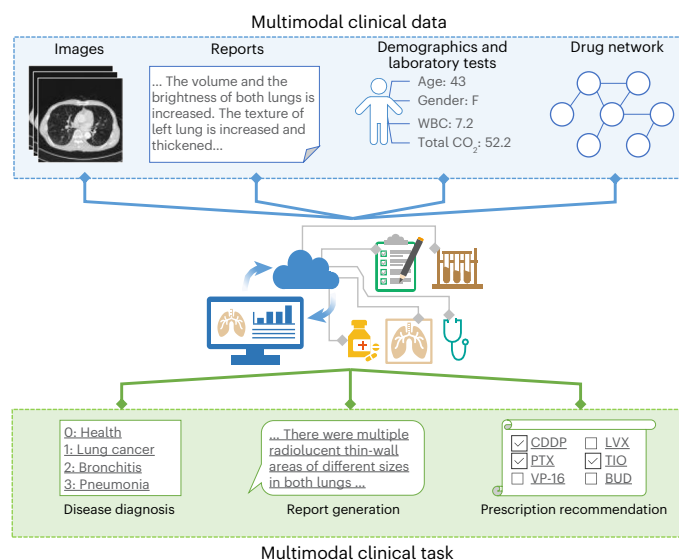


Fig. 1 | MedMPT learns from multimodal medical data within a unified framework. The multimodal knowledge (including, CT images, radiology reports, demographics, laboratory tests and drug relationships) enables MedMPT to perform diverse clinical tasks with different input and output modalities. WBC, white blood cell count; BUD, budesonide; CDDP, cisplatin; LVX, levofloxacin; PTX, paclitaxel; TIO, tiotropium; VP-16, etoposide.

representations without additional annotation^{17,18}. In MedMPT, we leverage the attention mechanism¹⁹ and extend self-supervised learning to multimodal medical data, capturing correlations within and across heterogeneous sources to extract representations that facilitate a wide range of downstream tasks. Specifically, we develop a multiscale and multimodal self-supervised approach tailored to the characteristics of medical data and clinical tasks, which performs contrastive and generative learning tasks to pretrain MedMPT on extensive pairs of CT images and free-text reports, equipping it with medical insights for comprehending multimodal data while alleviating the need for labour-intensive annotations. We investigate MedMPT on diverse clinically aligned tasks to comprehensively assess its performance. We evaluate disease diagnosis performance using chest CT data from the National Lung Screening Trial (NLST)²⁰ and MosMedData²¹ public datasets, and further collect a multimodal clinical dataset to assess MedMPT's performance on routine clinical tasks, including report generation and medication recommendation, based on relevant multimodal data. MedMPT consistently outperforms the comparison models, with significant improvements and better data and parameter efficiency, showing advantages across a broad spectrum of clinical tasks. The results validate the effectiveness of the multitask pretraining approach tailored to medical data, highlighting the significance of high-quality medical data and domain-specific knowledge in developing pretrained models for tasks closely aligned with real clinical demands. We investigate the clinical validity and potential impact of MedMPT on the routine work of radiologists through human–AI collaboration, indicating its potential to enhance both efficiency and healthcare quality. Qualitative analysis of interpretability in downstream tasks suggests that MedMPT can provide interpretable cues for decision-making, thereby enhancing the reliability and trustworthiness of medical AI models. MedMPT acquires holistic representations of multimodal medical data and benefits a wide range of downstream tasks within the clinical workflow (Fig. 1), making it more aligned with clinical practice and closer to the real sense of being general purpose. The human expert evaluations show the real-world applicability of MedMPT, indicating that human–AI collaboration is a promising approach for improving clinical workflows and ensuring high-quality patient care. Our findings reveal the potential of MedMPT

in the realm of medical applications and contribute valuable insights that will further the advancement of general-purpose AI within the medical domain.

Results

Overview

The aim of MedMPT is to pretrain the network to capture the multimodal knowledge in medical data and establish a fundamental basis for a wide range of subsequent medical tasks. To acquire medical knowledge from multimodal medical data without human annotation, we collected a large-scale multimodal dataset of 154,274 plain CT scans with corresponding radiology reports from The First Affiliated Hospital of Guangzhou Medical University at Liwan (FAHGMU-LW). We developed a multiscale and multimodal self-supervised learning method tailored to medical data and clinical applications, enabling MedMPT to acquire multiscale semantics across various modalities and enhance its performance in diverse clinically aligned tasks. MedMPT was initially pretrained using an institutional multimodal dataset. After pretraining, the pretrained parameters on MedMPT were transferred to three specific downstream tasks, that is, disease diagnosis, report creation and prescription recommendation, to initialize the parameters on the corresponding networks for the downstream tasks. We performed end-to-end fine-tuning using task-specific data as standard practice, updating the pretrained parameters with labelled data from the target downstream tasks to explore the effectiveness of MedMPT in diverse clinical tasks. The characteristics of the pretraining and downstream dataset are provided in Extended Data Table 1.

We compared MedMPT with publicly available vision–language pretrained models, including OpenAI CLIP²², PubMedCLIP²³, BiomedCLIP¹², Med-Flamingo²⁴ and LLaVA-Med²⁵, along with the commonly used foundational nature baseline and two MedMPT variants, MedMPT(CLIP) and MedMPT(2D), to evaluate the effectiveness of MedMPT (see Methods for details). For the comparative models designed specifically for two-dimensional (2D) images, we applied average pooling to aggregate the features of individual slices within a CT scan, obtaining a comprehensive representation of the entire scan. We investigated the effects of MedMPT under different transfer-learning strategies, which enhances the resource efficiency in clinical use and shows the adaptability and practical utility in clinical scenarios. The comprehensive assessments, including quantitative comparisons, qualitative analyses and human expert evaluations, show the effectiveness of MedMPT, indicating the advantages of the multiscale self-supervised pretraining approach tailored to medical data and clinical tasks, and highlighting the importance of high-quality domain-specific data for pretrained models to achieve clinically valid performance in medical practice.

MedMPT achieved notable improvement for lung disease diagnosis

We first explored the performance of MedMPT in CT-based lung disease diagnosis. Identifying lung disease through CT images is a fundamental task in clinical practice. We used the public dataset from NLST²⁰ for the task of lung cancer screening and MosMedData²¹ for the diagnosis and severity grading of coronavirus disease 2019 (COVID-19). The natural baseline was built upon the commonly used ViT²⁶ model pretrained with ImageNet-21k²⁷.

As shown in Fig. 2a, MedMPT consistently outperformed the compared models across various lung disease diagnosis tasks, achieving average accuracy improvements of 0.1139 over the natural baseline, 0.1072 over OpenAI CLIP, 0.1124 over PubMedCLIP, 0.0791 over BiomedCLIP, 0.0512 over Med-Flamingo and 0.0445 over LLaVA-Med. Specifically, for the task of lung cancer screening, MedMPT achieved an accuracy of 0.8958 (95% confidence interval (95% CI), 0.8953, 0.8981) and area under the receiver operating characteristic curve (AUROC)

of 0.9269 (95% CI, 0.9260, 0.9288), significantly outperforming the second-best LLaVA-Med, which achieved an accuracy of 0.8490 and an AUROC of 0.8423 ($P < 0.0001$). For identifying COVID-19, MedMPT performed significantly better than the comparative models, with accuracy of 0.8892 (95% CI, 0.8888, 0.8899) and AUROC of 0.9391 (95% CI, 0.9385, 0.9394). Even for the challenging task of COVID-19 severity grading, although overall performance was limited, MedMPT excelled by achieving an accuracy of 0.7477 (95% CI, 0.7477, 0.7493) and AUROC of 0.8521 (95% CI, 0.8513, 0.8527), yielding significant improvements of more than 0.0388 and 0.0530, respectively, over the comparative models. The relatively poor performance of the natural-baseline and OpenAI CLIP models exposed the inadequacy of relying solely on knowledge acquired from the general domain for precise medical image analysis. MedMPT showed superior performance across diverse pathological conditions, reflecting its generalizability and adaptability, which stem from its specialized techniques that integrate information across sequential slices to provide a holistic analysis of the entire CT scan. This capability is particularly beneficial for tasks such as lung cancer screening and severity grading of COVID-19, where it is crucial to capture the contextual relationships between slices holistically. This effectiveness is evidenced by the performance degradation observed between the complete MedMPT and its variants MedMPT(2D) and MedMPT(CLIP).

We further deepened our investigation by following the common practice of linear probing, comparing outcomes when the pretrained parameters of the vision encoder were frozen and only a linear prediction head was updated using labelled training data. In linear probe experiments, each comparison model is constrained to only 1,000–2,000 learnable parameters. The linear probing receiver operating characteristic (ROC) curves of all comparison models are shown in Fig. 2b. Although most comparison models performed poorly, the parameter-efficient MedMPT showed consistently impressive performance, achieving an AUROC of 0.8238 (95% CI, 0.8205, 0.8249) for lung cancer screening, 0.9175 (95% CI, 0.9171, 0.9182) for COVID-19 diagnosis and 0.7685 (95% CI, 0.7677, 0.7693) for COVID-19 severity grading. It is noteworthy that the parameter-efficient MedMPT performed on par with, or even surpassed, some end-to-end fine-tuned comparison models, despite having substantially fewer tuned parameters. The superior performance highlights the effectiveness and parameter efficiency of MedMPT. Detailed comparison results and statistical analyses are provided in Supplementary Tables 1 and 2.

MedMPT improved the semantic correctness for report generation

We evaluated the performance of MedMPT on the automatic generation of radiology reports from input CT scans, which enables MedMPT to enhance the efficiency and effectiveness of the clinical workflow. Reporting is an essential stage in the clinical workflow, serving as the conduit for conveying semantically rich information derived from medical images. It acts as a vital link in communication between radiologists and referring clinicians. A well-crafted radiology report should encapsulate comprehensive diagnostic imaging findings, articulating them in lucid and clinically pertinent medical language. Here we assessed the language quality and the semantic correctness of reporting with the commonly used metrics in image-captioning tasks, that is, the bilingual evaluation understudy (BLEU)²⁸, the recall-oriented understudy for gisting evaluation (ROUGE)²⁹ and the metric for evaluation of translation with explicit ordering (METEOR)³⁰. We observed that few existing vision–language pretrained models in the medical domain have explored the report generation task. Most models derived from the multimodal encoder structure of the Contrastive Language-Image Pretraining (CLIP) framework²², such as PubMedCLIP, BiomedCLIP and our MedMPT(CLIP), are incapable of handling generative tasks. This limitation highlights the necessity of developing a pretrained model specifically tailored for clinical tasks to meet

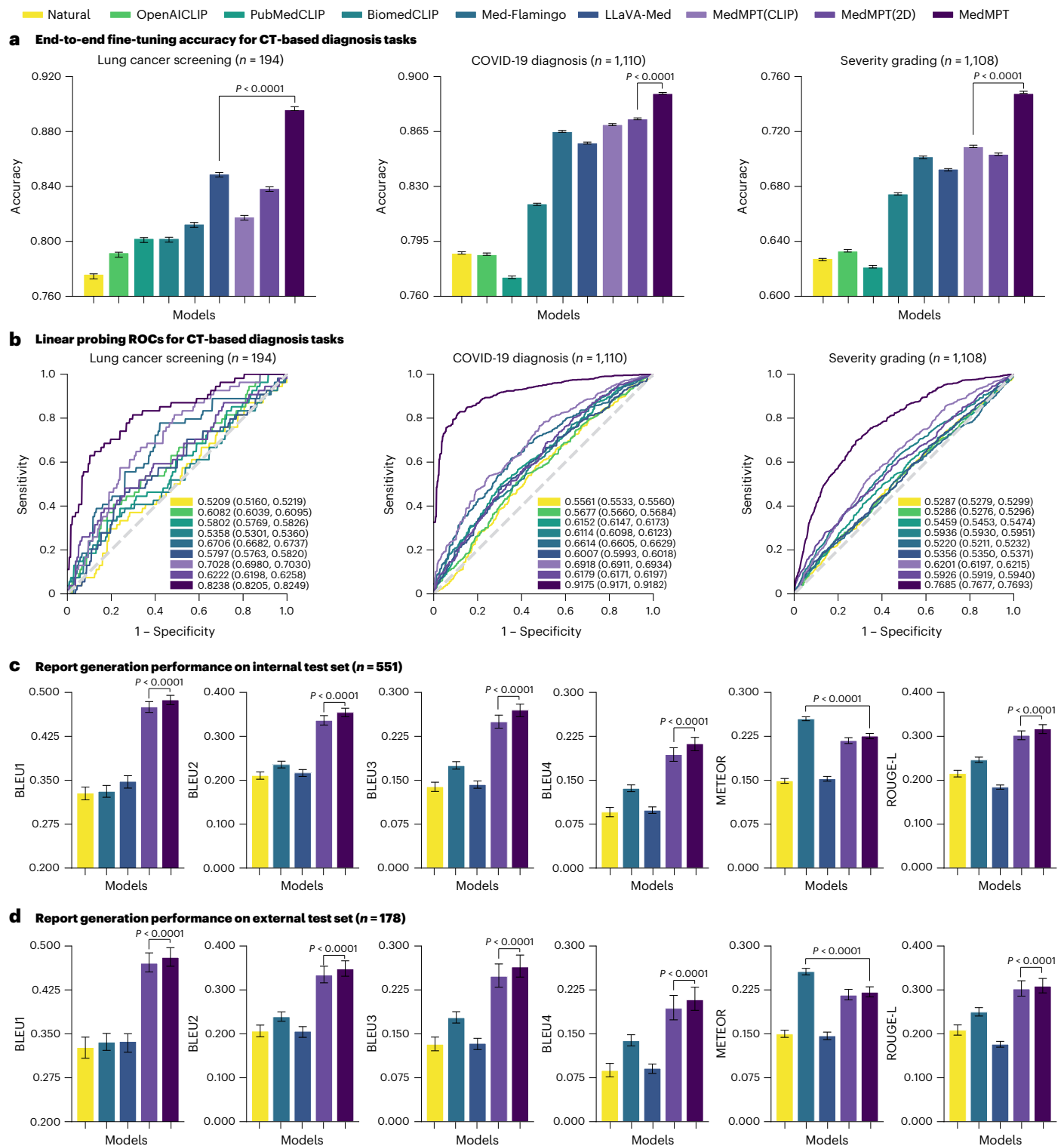


Fig. 2 | Disease diagnosis and report generation performance. **a**, Accuracy of end-to-end fine-tuning on CT-based diagnosis tasks: lung cancer screening ($n = 194$), COVID-19 diagnosis ($n = 1,110$) and severity grading of COVID-19 ($n = 1,108$). **b**, Linear probing ROC curves for the three diagnosis tasks. The values given in the plots are the area under the ROC curves with 95% CIs. The dashed line represents the performance of a random classifier for reference.

c, Quantitative metrics for report generation on the internal test set ($n = 551$). **d**, Report generation performance on the external test set ($n = 178$). Data are shown as means and 95% CIs (bootstrapped with 1,000 resamples). The P values are calculated using the two-sided Wilcoxon rank-sum test between MedMPT and the most competitive model. More statistical results can be found in Supplementary Tables 1–3.

the unique requirements of the medical field. We compared MedMPT with Med-Flamingo and LLaVA-Med in the report generation task. For the comparative models, we used instruction-guided generation as the standard practice. In addition, we developed a natural baseline

using a vision encoder from a pretrained ViT and a text decoder derived from BERT³¹.

As shown in Fig. 2c, MedMPT showed superior performance across most metrics, achieving a score of 0.4872 (95%CI, 0.4791, 0.4950)

for BLEU1, 0.3543 (95% CI, 0.3446, 0.3639) for BLEU2, 0.2696 (95% CI, 0.2585, 0.2799) for BLEU3, 0.2123 (95% CI, 0.2004, 0.2231) for BLEU4 and 0.3169 (95% CI, 0.3063, 0.3264) for ROUGE-L. Although the METEOR score of MedMPT (0.2254; 95% CI, 0.2208, 0.2299) was slightly lower than that of Med-Flamingo (0.2548; 95% CI, 0.2516, 0.2583), MedMPT showed significant advantages in the other metrics, outperforming the second-best model by more than 0.1391 in BLEU1, 0.1186 in BLEU2, 0.0947 in BLEU3, 0.0762 in BLEU4 and 0.0708 in ROUGE-L (all $P < 0.0001$). To verify the generalizability across centres, we conducted external validation using data collected from another institution. As shown in Fig. 2d, MedMPT showed robust performance, with scores of 0.4804 (95% CI, 0.4653, 0.4971) for BLEU1, 0.3477 (95% CI, 0.3312, 0.3661) for BLEU2, 0.2642 (95% CI, 0.2469, 0.2845) for BLEU3, 0.2081 (95% CI, 0.1901, 0.2296) for BLEU4, 0.2210 (95% CI, 0.2130, 0.2303) for METEOR and 0.3085 (95% CI, 0.2932, 0.3261) for ROUGE-L, showing only slight declines compared with the internal test set. Detailed results and statistical analyses are provided in Supplementary Table 3.

To further illustrate the performance of MedMPT and comparative models, we compared reports generated by MedMPT, Med-Flamingo and LLaVA-Med (Extended Data Fig. 1). MedMPT produced more accurate and professional lung pathology reports, showing close alignment with human-written reports. In contrast, the reports generated by comparison models often included irrelevant or hallucinated contents. The suboptimal performance of these models can be primarily attributed to a misalignment between their pretraining tasks and clinical report generation. Radiology reports require detailed descriptions that link specific visual patterns in CT scans to medical terminology, which is essential for accurately understanding medical conditions and generating radiologist-like descriptions. However, most existing models, originally designed as biomedical chatbots, were pretrained on more generalized and freestyle image-caption pairs. The advantages of MedMPT highlight the importance of domain-specific data and task alignment in developing foundational models for clinical applications.

We also performed zero-shot evaluation, in which the pretrained models were directly used to generate imaging reports from the input CT scans without any downstream data fine-tuning, to assess the generalizability of pretrained models. In zero-shot evaluations, MedMPT achieved a BLEU4 score of 0.1961 (95% CI, 0.1837, 0.2089) and a ROUGE-L score of 0.3023 (95% CI, 0.2917, 0.3132), with decreases of less than 0.02 compared with its fully supervised performance. In contrast, the comparison models showed substantial performance degradation, highlighting the robustness of MedMPT. Detailed zero-shot results and statistical analyses are provided in Supplementary Table 4.

MedMPT performed expert-like prescription recommendations

To further validate the models in clinically relevant applications, we evaluated their performance in predicting medications based on multimodal examination data. Prescribing medications is a critical and complex component of the clinical workflow, necessitating a thorough examination of both patient conditions and medication interactions to guarantee appropriate recommendations. Specifically, we compiled a multimodal prescription recommendation dataset in which each patient visit included CT images, radiology reports, demographic information, laboratory test results and a list of medications that had been retrospectively recommended, to evaluate prescription recommendation performance. The natural baseline used the ImageNet-pretrained ViT and pretrained BERT as the vision and text encoder, respectively.

Figure 3 presents the precision, recall and *F1* score (the harmonic mean of precision and recall) metrics of comparative models, using a confidence level (also known as cut-off value) of 0.5 for each medication. As shown in Fig. 3a, MedMPT achieved a precision of 0.5703 (95% CI, 0.5695, 0.5715), recall of 0.4438 (95% CI, 0.4426, 0.4442) and *F1* score of 0.4576 (95% CI, 0.4565, 0.4580) with end-to-end fine-tuning.

MedMPT significantly outperformed the second-best models, surpassing PubMedCLIP in precision (0.5532; 95% CI, 0.5517, 0.5536) and BiomedCLIP in recall (0.4140; 95% CI, 0.4127, 0.4144) and *F1* score (0.4195; 95% CI, 0.4182, 0.4198) (all $P < 0.0001$). We also examined performance under different settings by fixing the number of recommended medications and adjusting the predicted confidence thresholds (Supplementary Fig. 1), with MedMPT consistently showing superior performance over comparative models. The external validation performance is shown in Fig. 3b. MedMPT achieved a precision of 0.5349 (95% CI, 0.5338, 0.5372), recall of 0.4036 (95% CI, 0.4019, 0.4048) and *F1* score of 0.4262 (95% CI, 0.4249, 0.4276). Despite potential variability and distribution shifts, MedMPT maintained stable performance, outperforming the comparison models with well-balanced precision and recall. Detailed results and statistical analyses are provided in Supplementary Tables 5 and 6.

Considering the complexity of medical prescriptions, where different clinicians may recommend various medications for the same visit, we engaged four junior and two senior clinicians to evaluate the prescriptions recommended by MedMPT. The test data were randomly divided into two groups, each of which was assessed by two junior clinicians and confirmed by a senior clinician. The junior clinicians were asked to check and score the prescriptions based only on the patient examination results, that is, the CT scan, radiology report and laboratory tests, pointing out any substance misuse or errors that might be harmful due to drug–drug interactions (DDIs) and identifying any missing medications. The retrospective prescription labels were inaccessible to all clinicians. The detailed scoring criteria are explained in the Methods. Overall, about 73.9% of the prescriptions received a perfect score of 5 on a 6-point scale (ranging from 0 to 5, where higher scores indicate better performance), which means that MedMPT predicted all the medications needed by the patient with no harmful DDIs in the prescriptions. The average score for all evaluated prescriptions was 4.54 out of 5, indicating that most prescriptions successfully anticipated the patients' most necessary medications. Figure 3c shows two examples of the prescriptions generated by MedMPT. Given the multimodal data, MedMPT achieved high recall and precision values, with false positives often serving similar purposes as false negatives and being closely related to the diagnosis. The results showed that MedMPT can provide reasonable, expert-like prescriptions based on patient examinations and medication associations. The clinicians recognized that MedMPT offers valuable advice and assistance in prescribing, enhancing their decision-making abilities and efficiency.

We further evaluated MedMPT using a parameter-efficient strategy, keeping the pretrained parameters frozen and only updating the additional task-specific module. As shown in Fig. 3d, MedMPT showed robust performance, achieving a precision of 0.5744 (95% CI, 0.5735, 0.5755), recall of 0.4333 (95% CI, 0.4321, 0.4338) and *F1* score of 0.4502 (95% CI: 0.4491, 0.4507), similar to comparative models with more learnable pretrained parameters. The evaluation across different transfer strategies, data distributions, metrics and comparison models highlights the robustness and efficiency of MedMPT in leveraging pretrained knowledge for downstream tasks, along with its adaptability in clinical applications due to its modular architecture. Detailed results of the parameter-efficient fine-tuning and statistical analyses are provided in Supplementary Table 7.

MedMPT enhanced routine efficiency and healthcare quality in human–AI collaboration

While MedMPT showed outstanding quantitative performance across several clinical tasks, it is essential to further explore its clinical validity and potential impact on routine work. Given the importance of accurate and timely radiology reports in clinical practice, we investigated how MedMPT can enhance these aspects of radiologists' work in report generation. To assess the clinical validity of MedMPT, we conducted human–AI collaboration experiments to analyse its potential

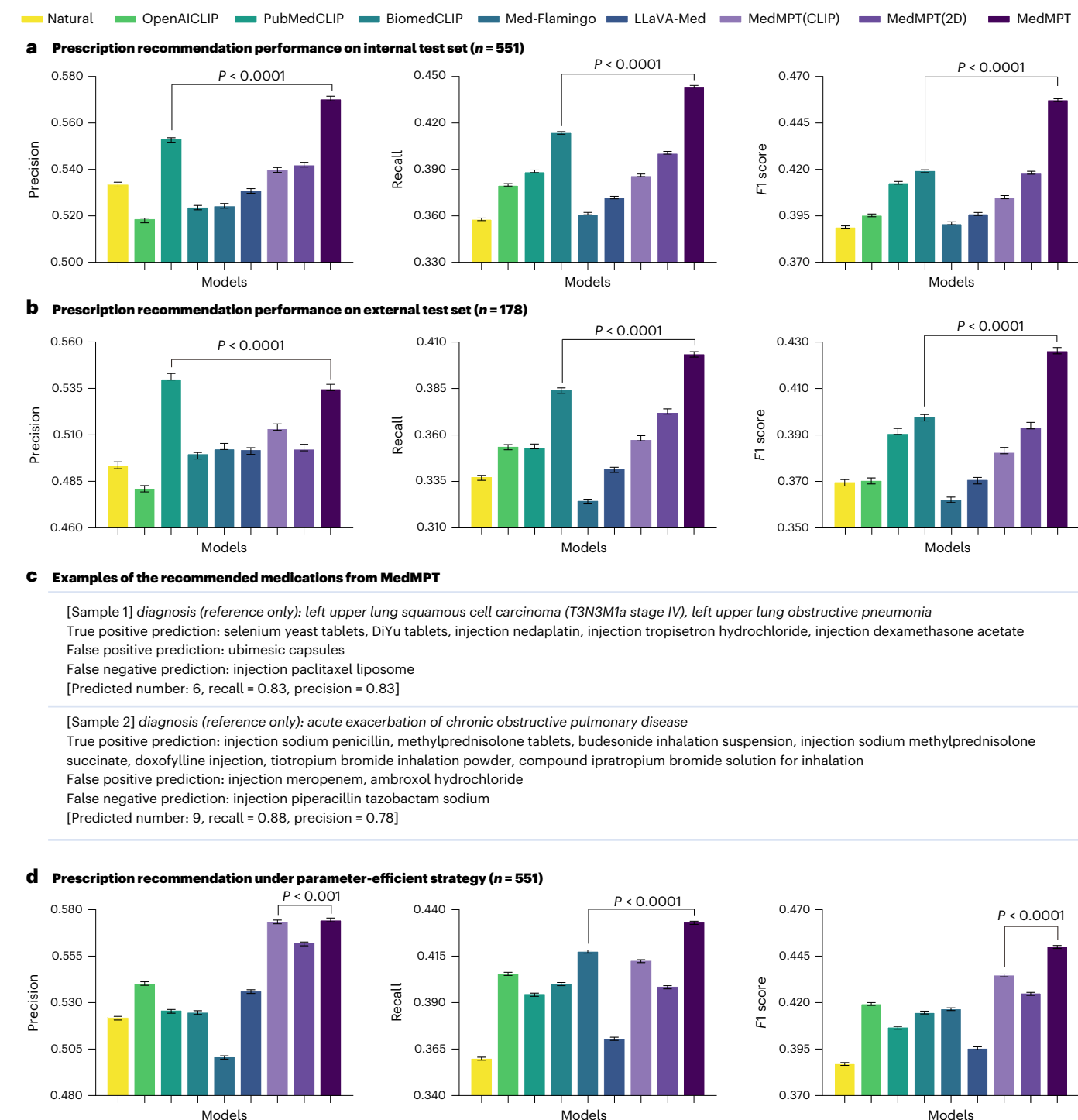


Fig. 3 | Performance on multimodal prescription recommendation. a, Recall, precision and F1 score of MedMPT and comparison models on the internal test set ($n = 551$) under end-to-end fine-tuning. **b,** Performance comparison on the external test set ($n = 178$). **c,** Two examples of the recommended medications from MedMPT compared with the retrospective prescription labels. Patient diagnoses are shown for illustrative purposes only and were not accessed by the model. Diyu tablet (Diyu Shengbai Pian), a traditional Chinese medicine used

to increase white blood cell count. **d,** Performance on the internal test set using a parameter-efficient strategy in which pretrained weights were frozen and only task-specific modules were fine-tuned. Data are shown as means and 95% CIs (bootstrapped with 1,000 resamples). The P values are calculated using the two-sided Wilcoxon rank-sum test between MedMPT and the most competitive model. More statistical results can be found in Supplementary Tables 5–7.

to help radiologists improve both the efficiency and quality of their reporting workflows.

First, we compared the time required for radiologists to complete imaging reports with and without model assistance. We engaged 2 junior (less than 5 years of experience), 2 mid-level (5–10 years of experience), and 2 senior (over 10 years of experience) radiologists

and recorded the time they spent completing each report, to investigate the impact of MedMPT on clinical efficiency across different experience levels. Figure 4a shows the completion times from the human–AI collaboration experiment, which show that the integration of MedMPT consistently improves efficiency across all experience levels. Specifically, with MedMPT assistance, the average reporting

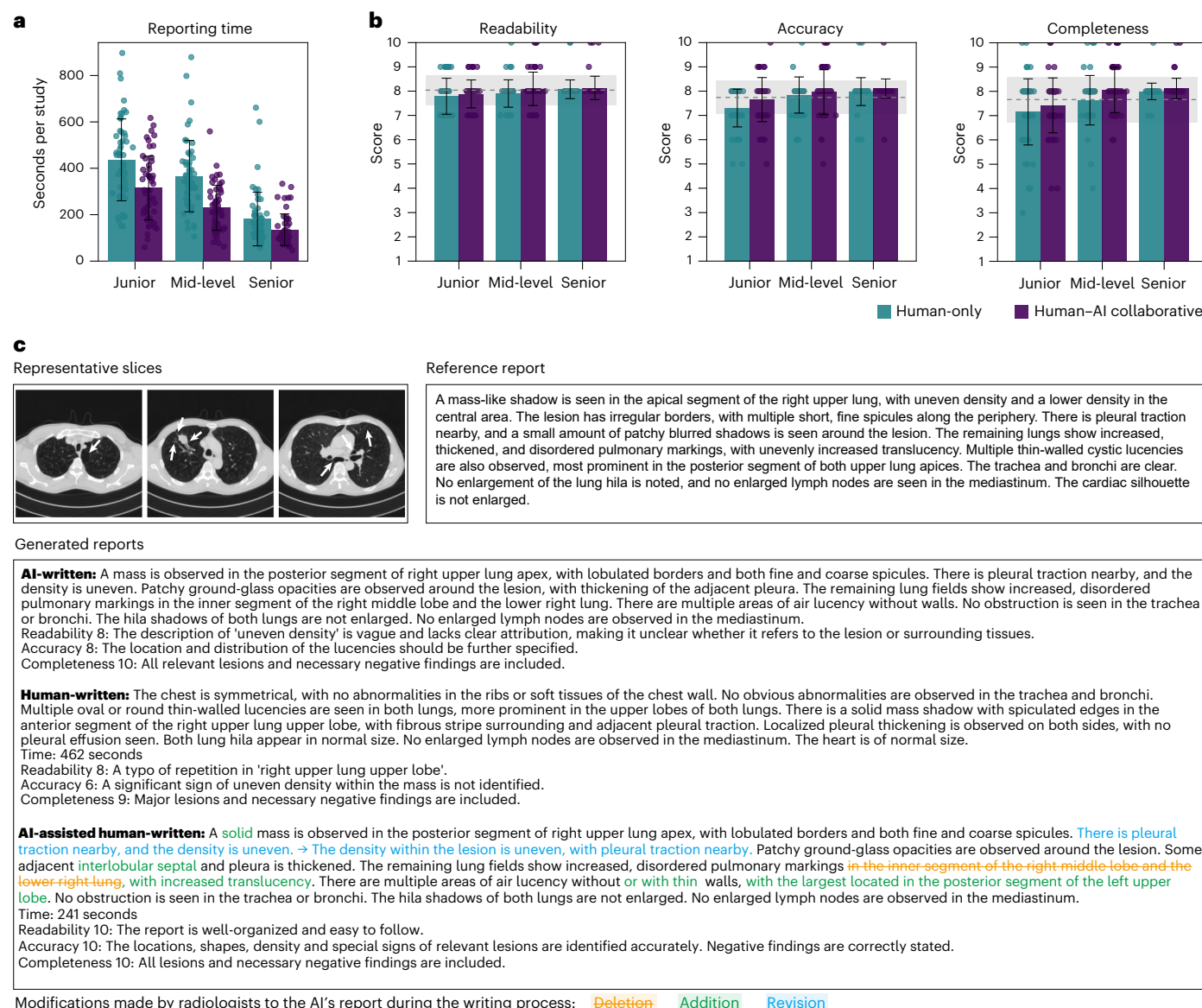


Fig. 4 | Human–AI collaboration in radiology report generation. Comparison of reporting time and report quality with and without AI assistance following the routine radiology workflow. **a**, Reporting time of radiologists with and without access to MedMPT-generated reports as references. **b**, Quality scores of the reports in terms of readability, accuracy and completeness. Bars show the mean values with error bars representing the standard deviation. Results of individual data points are overlaid as dots. Grey dashed lines and shading show the mean and standard deviation of the AI-generated report scores. **c**, Case study of human–AI collaboration in report generation. Shown is the reference report,

the most relevant CT slice(s) highlighting key findings (typically abnormal lesions), and the reports generated by MedMPT (AI-written), a radiologist alone (human-written) and a radiologist with AI assistance (AI-assisted human-written). The revisions made by the radiologist are highlighted using colour-coded annotations. The completion times and quality scores for the reports generated by radiologists are included. On the basis of the MedMPT-generated report, the human–AI collaborative report combined the clinical insights of the radiologist with the details generated by MedMPT, improving its clinical relevance and applicability within a shorter time frame.

time decreased from 334.47 s to 228.28 s across all radiologists, indicating a time reduction of 31.75%. Senior radiologists benefited the most (37.34% reduction), followed by mid-level (36.52%) and junior radiologists (24.85%), showing an inverse correlation between efficiency gains and experience level. We examined revisions made by radiologists to the generated reports and observed that this inverse correlation might be because less-experienced radiologists tended to follow a more rigid and standardized workflow, making it difficult for them to integrate and adapt the AI-generated reports seamlessly, leading to extra time spent on verification and reformatting. In contrast, experienced radiologists showed greater adaptability. They quickly assessed the adequacy of the AI-generated content and focused their revisions on clinical content rather than stylistic preferences,

allowing for more efficient integration of the AI-generated report. These observations suggest that experienced radiologists are more proficient in using AI tools, and emphasize the need for additional training to help less-experienced radiologists integrate AI more effectively into their workflows. The detailed reporting times of individual radiologists are provided in Supplementary Table 8.

To quantitatively assess the quality of the reports generated with or without human–AI collaboration, we engaged a senior radiologist (over 10 years of experience) to review and score the generated reports. The radiologist was blinded to the origin of the reports and evaluated them based on three key clinical criteria: readability, accuracy and completeness. Reports were scored on a scale from 1 to 10, with higher scores indicating better performance. The evaluation

criteria closely aligned with the clinical requirements. Specifically, readability assessed the quality of the writing, including grammatical correctness, logical flow and clarity. Accuracy measured how well the report reflected the patient's actual condition, ensuring that essential clinical details were included. Completeness evaluated whether all necessary observations were made to support an accurate clinical diagnosis. The detailed scoring criteria are provided in Methods.

Figure 4b presents the report quality scores from the human–AI collaboration experiment. Overall, collaboration with MedMPT is beneficial for improving report quality, especially with regards to accuracy and completeness for less experienced radiologists. We quantified both the critical error rates, defined as reports scoring 4 or below in readability, accuracy or completeness, and clinical acceptance rates, defined as reports scoring 7 or above in all 3 evaluation aspects and deemed suitable for clinical use with minimal revision. Among reports written independently by radiologists, 6% of those produced by junior radiologists contained critical errors. This rate dropped to 2% for mid-level radiologists, with no critical errors found in reports from the senior group. In comparison, 4% of the AI-generated reports contained critical errors, mainly due to incomplete clinical details, placing the AI-generated reports between those of the junior and mid-level radiologists. In contrast, for human–AI collaboration, the critical error rate of the junior group reduced to 2%, and no critical errors were observed with the mid-level or senior radiologists. In terms of clinical acceptance, 73% of AI-generated reports met the clinical standard, whereas the radiologist-written reports showed more variability, with 67%, 88% and 96% for junior, mid-level and senior radiologists, respectively. Collaboration with MedMPT further improved these rates to 75% for the junior group, 92% for the mid-level group and 98% for the senior group. These results highlight that MedMPT has great potential to enhance the quality of radiological reports, especially for less experienced radiologists. The reports generated by MedMPT showed strong baseline performance to support radiologists in producing more accurate, complete and clinically acceptable reports. Human–AI collaboration not only reduces the need for major revisions but also narrows the gap across experience levels.

In discussions with participating radiologists, we next examined the strengths and weaknesses of the reports. MedMPT reliably detected primary lesions and identified clinically relevant features using professional terminology, particularly for conditions such as emphysema, chronic obstructive pulmonary disease and interstitial pneumonia. Although minor inaccuracies or occasional lack of clinical focus were noted, they did not substantially affect the clinical utility of the reports. The performance of radiologists varied by experience: senior radiologists tended to be more succinct and focus on clinically relevant lesions, sometimes providing less detail; junior radiologists typically offered more detailed descriptions but were more prone to omissions. In contrast, collaborative reports leveraged the strengths of both AI and human expertise, benefiting from the comprehensive observations provided by MedMPT and the clinical insights of the radiologists, and achieving a higher quality of assessment. Figure 4c provides an example generated by MedMPT (AI-written), written independently by radiologists (human-written), and produced through collaboration between MedMPT and radiologists (AI-assisted human-written), along with their scores and evaluation rationale. The reports generated by MedMPT provided a strong foundation for radiologists, allowing them to identify the lesions rapidly and refine the descriptions easily, speeding up report generation and enhancing the accuracy and completeness of the final reports. More collaborative examples can be found in Extended Data Figs. 2 and 3.

In conclusion, the evaluation procedure simulating routine radiological workflows provides insights into the practical impact of MedMPT on the daily work of radiologists. Our study shows the

potential of MedMPT to enhance both the efficiency and quality of clinical workflows across experience levels through effective human–AI collaboration. The results suggest that a good use for AI is as a supportive tool to augment, rather than replace, healthcare professionals. These findings highlight the importance of further exploring human–AI collaboration, particularly in optimizing AI use to maximize clinical utility and in strengthening training for clinicians on AI integration.

MedMPT provided explainable evidence for clinical decisions

Figure 5 presents several cases related to lung cancer diagnosis, radiology report generation and prescription recommendation with MedMPT. Figure 5a show the attention map of the CT slice for the cases with lung cancer. MedMPT effectively identified discriminative features within the image and assigned higher weights to these crucial locations when making decisions. The highlighted regions provide explainable evidence and essential clues for clinical decisions. Figure 5b shows two cross-attention maps of visits with lung abnormalities. We showed the saliency maps for the highlighted keywords in the reports of the slices within a CT scan and displayed the attention maps for the two slices with the highest probabilities. For example, in the first case, the word 'reticular' is primarily associated with some diffuse lesions, so the distribution of slice saliency is relatively smooth and the slices with the highest weights contain the region with a high incidence of lung cancer for the whole scan. Inside the slice, MedMPT showed high attention to the area with reticular shadows. Similarly, for the second case, which was observed to have a localized pulmonary nodule, MedMPT accurately focused on the very few slices with a sharp weight distribution of the slices and correctly concentrated on the area surrounding the nodule lesion within the slice. The attention distribution indicates that MedMPT accurately identified and focused on the relevant regions of interest within the lung scan, and effectively aligned textual information with specific anatomical regions and abnormalities. Figure 5c provides the attention map with respect to the sample medications prescribed. MedMPT allocated higher attention to the specific regions corresponding to the signs targeted by the medication in the image. More visualization examples are provided in Supplementary Figs. 2–4. The results suggest that MedMPT has the potential to enhance the interpretability and clinical utility of its predictions, thereby enabling healthcare professionals to make informed decisions based on reliable and clinically relevant information. To further substantiate these qualitative insights, additional quantitative evaluations are necessary in future studies, and careful consideration should be given to how the diversity of reports and medications is interpreted in these evaluations to accurately reflect clinical needs.

Real-world data hold valuable information beyond one-to-one mapping label supervision

Another prominent finding from the prescribing experiments is that real-world datasets inherently capture prior knowledge that benefits clinical tasks. Specifically, we observed that the co-occurrence patterns of medications in the dataset provide valuable information beyond the traditional one-to-one label supervision. We quantified these associations by extracting a co-occurrence matrix (Supplementary Fig. 5) from the training set and 17,060 additional prescriptions, which were retrieved from the picture archiving and communication system (PACS) but excluded during the construction training-test set due to incomplete inspection modalities. This matrix measures how frequently each pair of the target medications appears together in the same prescription. From this co-occurrence matrix, we derived a label co-occurrence graph, representing the dependencies between different medications. Considering the complexity of prescribing, we pruned pairs with low co-occurrence probabilities (less than 0.5) to reduce noise from weak or spurious connections. The remaining co-occurring patterns (Fig. 6a) provide valuable prior information

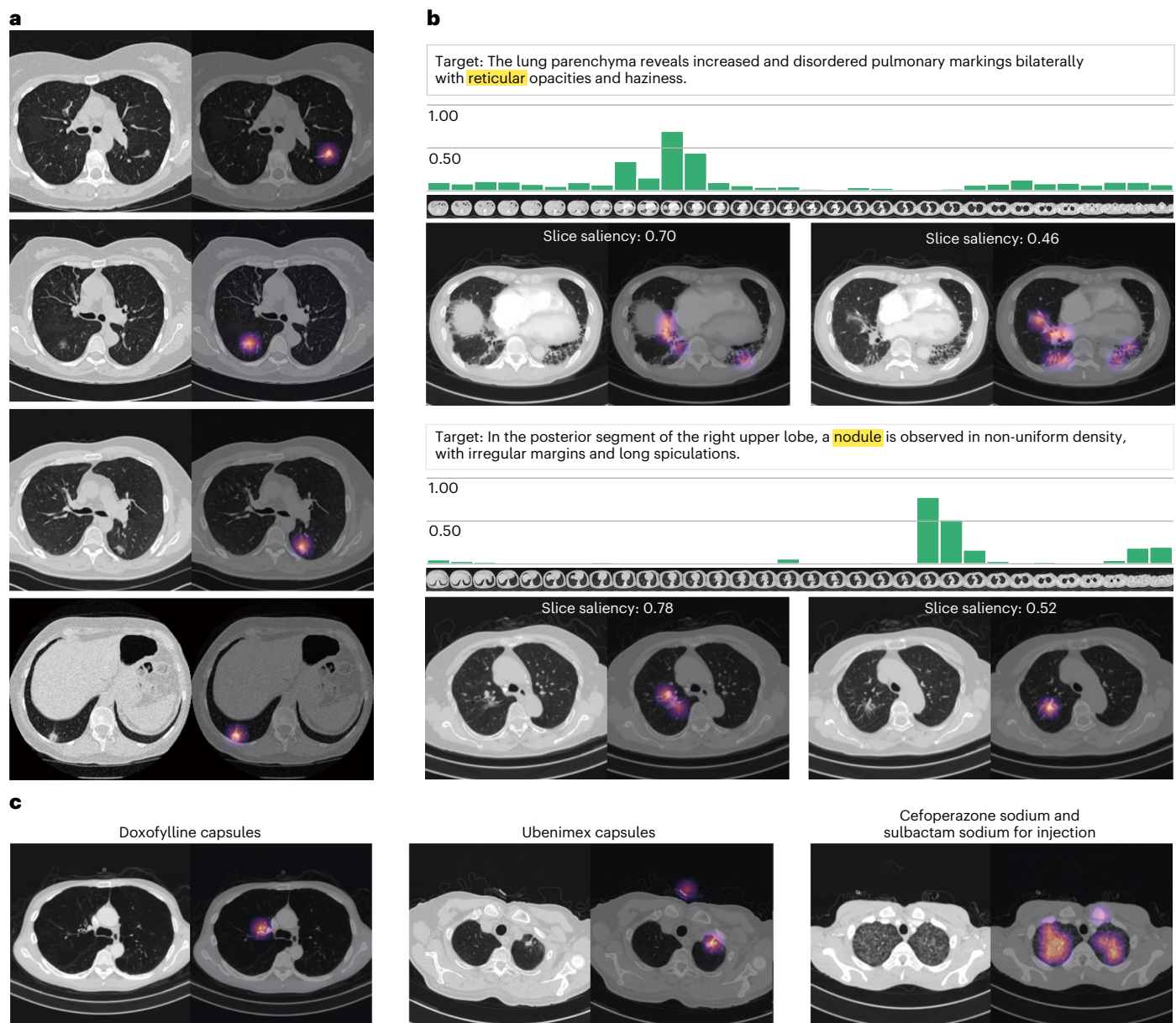


Fig. 5 | Qualitative evaluation of MedMPT. a, The CT slices (left) and corresponding attention maps generated by MedMPT (right) for the cases with lung cancer. MedMPT allocated higher attention weights (light regions within the attention maps) to areas closely associated with tumour lesions, correctly identifying critical regions of interest within the CT images. **b,** Examples of the report generated by MedMPT (top), slice weights corresponding to the highlighted keywords (middle), and attention maps within the two slices

receiving the highest weights (bottom). The values represent the normalized attention weights, where higher values indicate greater model attention. MedMPT accurately focused on the slices containing the related lesion within a scan and identified the relevant regions of interest within the slices. **c,** The original CT images (left) and corresponding attention maps (right) of MedMPT towards certain medications.

on medication relationships for prescription recommendations. For example, although both injection dexamethasone acetate and injection methylprednisolone sodium succinate share common characteristics as corticosteroids with anti-inflammatory properties, injection dexamethasone acetate shows a notable co-occurrence probability with medications such as injection tropisetron hydrochloride, ubenimex capsules and selenium yeast tablets. In contrast, injection methylprednisolone sodium succinate is commonly prescribed alongside medications such as ambroxol hydrochloride, budesonide suspension and compound ipratropium bromide solution for inhalation. These co-occurrence patterns suggest that dexamethasone acetate injections are potentially associated with specific medical conditions requiring anti-inflammatory or immunosuppressive effects, whereas

methylprednisolone sodium succinate injections appear more strongly linked to respiratory conditions, particularly those involving inflammation of the respiratory system.

Inspired by this observation, we constructed a medication co-occurrence graph to represent these associations, leveraging the clinical relevance of medication relationships to improve prescription accuracy and better align with clinical practice. The detailed adaptation algorithm is described in Methods. We explored the effects of integrating the medication co-occurrence graph with multimodal data and observed that removing the co-occurrence graph resulted in a performance decline in precision by 0.0215, recall by 0.0034 and *F1* score by 0.0081. By incorporating the medication co-occurrence graph, MedMPT was able to learn beyond individual label supervision

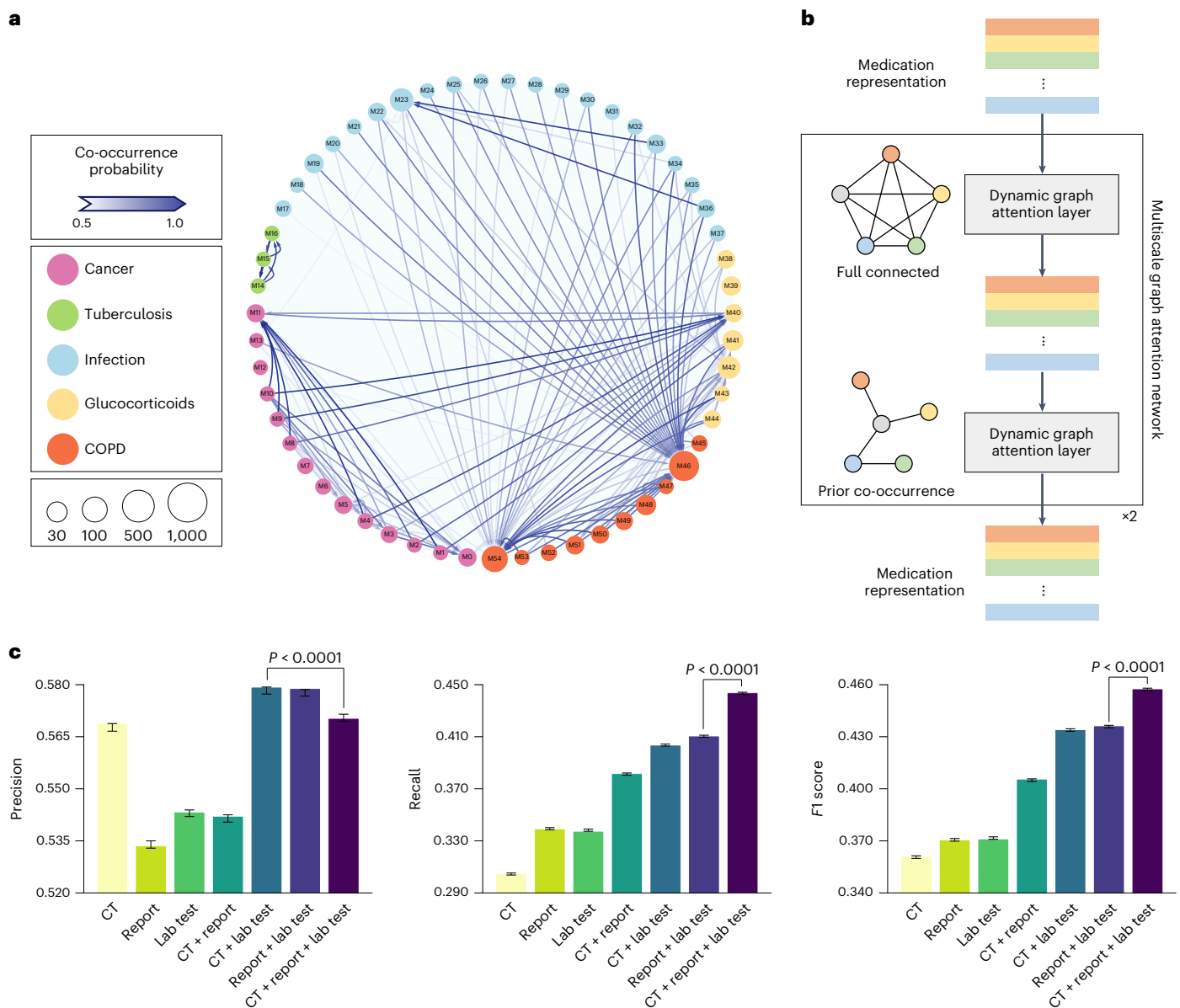


Fig. 6 | Overview of the medication modeling framework and its performance analysis. a, The co-occurrence patterns of the medications, which capture reasonable associations and dependencies between different medications. The numbers (30, 300, 500 and 1,000) indicate reference node sizes, representing the number of samples for each medication. The full names of medications M0–M54 are listed in the Supplementary Table 19. **b,** The framework of the medication encoder. A multiscale graph attention network is used to capture the relationship between medications, facilitating comprehensive data integration and fostering intricate interactions while retaining prior information and promoting essential medication patterns. **c,** Ablation analysis

of the impact of multimodal inputs on the performance of prescription recommendation (internal test set, $n = 551$ samples). Data are shown as means and 95% CIs (bootstrapped with 1,000 resamples). The P values are calculated with two-sided Wilcoxon rank-sum tests between the full modalities model and the most competitive model. The results showed that incorporating additional input modalities significantly enhanced the comprehensive analysis of the patient's condition, consistently facilitating prediction performance. The model using the full set of modalities achieved the highest $F1$ score, indicating a well-balanced performance in terms of both precision and recall. More statistical results can be found in Supplementary Table 9.

and take advantage of the collective knowledge contained within the dataset to provide more robust and reliable results for complex clinical tasks.

Multimodal data play an essential role in clinical applications

To assess the individual impact of each input modality and show the importance of multimodal data in medical tasks, we conducted a thorough ablation study where we systematically added one modality at a time as input for prescription recommendations. The results (shown in Fig. 6c) revealed the significance of incorporating multimodal input for complex clinical applications. By leveraging the complete

set of input modalities, including CT images, reports, and the structured data of demographic information and laboratory test results, MedMPT achieved superior performance in the prescription recommendation. While the highest recall and $F1$ scores with a single modality were only 0.3392 (95% CI, 0.3385, 0.3401) and 0.3711 (95% CI, 0.3709, 0.3723), respectively, the inclusion of input modalities consistently improved these metrics. Moreover, the consistent performance across metrics showed the reduced variability and enhanced robustness of models with multiple input modalities. The results of the modality ablation study emphasize the holistic nature of medical data and highlight the significance and necessity of incorporating diverse

Table 1 | Ablation study of MedMPT across different tasks

Ablation method	Downstream tasks (evaluation metrics), mean (95% CI)				
	Cancer screening (AUROC)	COVID-19 identification (AUROC)	Severity grading (AUROC)	Report generation (BLEU4)	Prescription recommendation (F1 score)
Without MIR	0.9053 (0.9044, 0.9075)	0.9335 (0.9330, 0.9341)	0.8439 (0.9432, 0.8447)	0.2006 (0.1890, 0.2126)	0.4440 (0.4429, 0.4445)
Without CMG	0.9098 (0.9085, 0.9116)	0.9378 (0.9371, 0.9381)	0.8495 (0.8492, 0.8506)	0.2073 (0.1954, 0.2188)	0.4434 (0.4422, 0.4438)
Without ATF	0.8948 (0.8937, 0.8973)	0.9333 (0.9326, 0.9336)	0.8452 (0.8448, 0.8463)	0.2038 (0.1929, 0.2146)	0.4388 (0.4375, 0.4390)
Without IIC-slice	0.9043 (0.9036, 0.9067)	0.9331 (0.9324, 0.9333)	0.8443 (0.8441, 0.8455)	0.2091 (0.1981, 0.2213)	0.4392 (0.4380, 0.4395)
Without IIC-scan	0.9061 (0.9047, 0.9077)	0.9132 (0.9127, 0.9140)	0.8416 (0.8413, 0.8428)	0.2015 (0.1899, 0.2132)	0.4435 (0.4425, 0.4441)
Without ITC	0.8048 (0.8029, 0.8074)	0.9202 (0.9196, 0.9207)	0.8200 (0.8195, 0.8210)	0.1922 (0.1804, 0.2033)	0.4273 (0.4258, 0.4274)
MedMPT	0.9269 (0.9260, 0.9288)	0.9391 (0.9385, 0.9394)	0.8521 (0.8513, 0.8527)	0.2123 (0.2004, 0.2231)	0.4576 (0.4565, 0.4580)

The performance of the model and its ablation variants is summarized on several tasks, each evaluated with task-specific metrics as specified. Values are presented as mean with 95% CIs (bootstrapped with 1,000 resamples). The last row represents the full MedMPT, and each of the preceding rows represents the removal of one of the key modules or subtasks from the complete MedMPT during pretraining. More detailed evaluation metrics and statistical results can be found in Supplementary Tables 10–12. ATF, attention-based slice fusion; CMG, cross-modal generation; IIC-slice and IIC-scan, slice- and scan-level image–image contrastive learning; ITC, image-text contrastive learning; MIR, masked image reconstruction.

modalities of medical data to develop medical models and provide robust and accurate clinical decision support in medical applications. More detailed comparisons and statistical analyses can be found in Supplementary Table 9.

MedMPT showed unified adaptability across clinical applications

We further explored a multi-task fine-tuning strategy to evaluate the effectiveness of MedMPT in addressing multiple clinical tasks simultaneously, which would be helpful to enhance the resource efficiency and adaptability of MedMPT across various clinical applications. We evaluated and compared the performance of the joint model with task-specific models in cancer screening, report generation and prescription recommendation tasks (Extended Data Table 2 and Supplementary Tables 13–15).

As shown in Extended Data Table 2, compared with task-specific models, the joint model showed markedly higher parameter efficiency while achieving better generalizability across tasks and datasets, making it suitable for complex, resource-constrained clinical environments. Specifically, on the internal test set, task-specific models generally achieved higher performance than the joint model on individual tasks. For example, in the report generation task, the task-specific model achieved a BLEU4 of 0.2123 compared with 0.1943 for the joint model on the internal set. In the prescription recommendation task, the task-specific model reached an *F1* score of 0.4576 compared with 0.4488 for the joint model. However, the joint model showed better generalizability on external data. In report generation, the joint model slightly improved its BLEU4 score to 0.1957 on the external set, whereas the task-specific model dropped to 0.2081. Similarly, in prescription recommendation, the *F1* score of the joint model decreased modestly to 0.4370, whereas the task-specific model declined more notably to 0.4262. Comprehensive evaluation metrics are provided in Supplementary Tables 13–15.

These results indicate that while task-specific models achieve slightly higher performance on internal data, the joint model shows superior generalizability under distribution shifts, benefiting from the diversity of the multitask data and the knowledge transfer across tasks to learn more robust and generalized features. Notably, the joint model achieved this with less than 58% of the parameters required by task-specific models, showing its efficiency in both performance and resource consumption. As a promising future direction, it would be valuable to explore more advanced strategies, such as mixture-of-experts³² and adapter³³ strategies, to further balance the task-specific performance with the robust generalization capabilities of the joint model, aiming to fully leverage the strengths of both approaches in diverse clinical scenarios.

All self-supervised modules contribute to MedMPT

We conducted a comprehensive ablation study to assess the impact of each module within the proposed multiscale self-supervised pre-training approach. As summarized in Table 1, the full MedMPT model consistently outperformed its ablated variants across all tasks. The most pronounced effect was observed upon the exclusion of the image–text contrastive component, which resulted in the largest reductions. Specifically, the removal of the image–text contrastive component led to a notable decline in performance, with AUROC in the CT-based lung cancer screening task dropping by 0.1221 and the *F1* score in the prescription recommendation task decreasing by 0.0303. This degradation reveals the essential role of intermodal interactions in aligning textual descriptions with corresponding visual data, which is critical for achieving accuracy and relevance in clinical outputs. Removing the attention-based slice fusion also resulted in notable performance declines, particularly in tasks that depend on global understanding, such as a drop of 0.0321 in lung cancer screening and 0.0085 in report generation, underscoring the importance of capturing interslice relationships and extracting a comprehensive understanding of entire CT scans in complex clinical tasks. Overall, these modules contributed to the effectiveness of MedMPT across diverse clinical tasks. Detailed ablation results can be found in Supplementary Tables 10–12.

Discussion

Recently, large pretrained models supporting a wide range of downstream tasks have attracted increasing interest in various domains. In the medical domain, several studies have focused on developing language models^{8,14,34–40} and vision models^{9,10,41–45} to handle structured and unstructured medical data for various applications. Recent studies^{46,47} have introduced innovative algorithms to develop foundation models for biomedical imaging, which enable efficient model training and reduce reliance on large-scale labelled data. However, medical data inherently contain multiple modalities. There is a wide range of tasks that require comprehensive analyses of patients' conditions to perform accurate management. Exploring each modality separately is inadequate and suboptimal for most medical applications. There have been attempts to leverage multimodal pretrained models in the medical domain^{11,12,23–25,48–50}. However, existing vision–language models primarily focus on vision–language processing and show limited support for clinically aligned tasks. Consequently, these models fall short of meeting the requirements for clinical usage. The clinical validity of the multimodal pretrained models for medicine remains underexplored.

We observed that a few recent studies^{51,52} have started to explore multimodal biomedical foundation models across a broad range of clinically aligned tasks with human evaluations. However, there is still a lack of investigation and evidence on how to effectively integrate

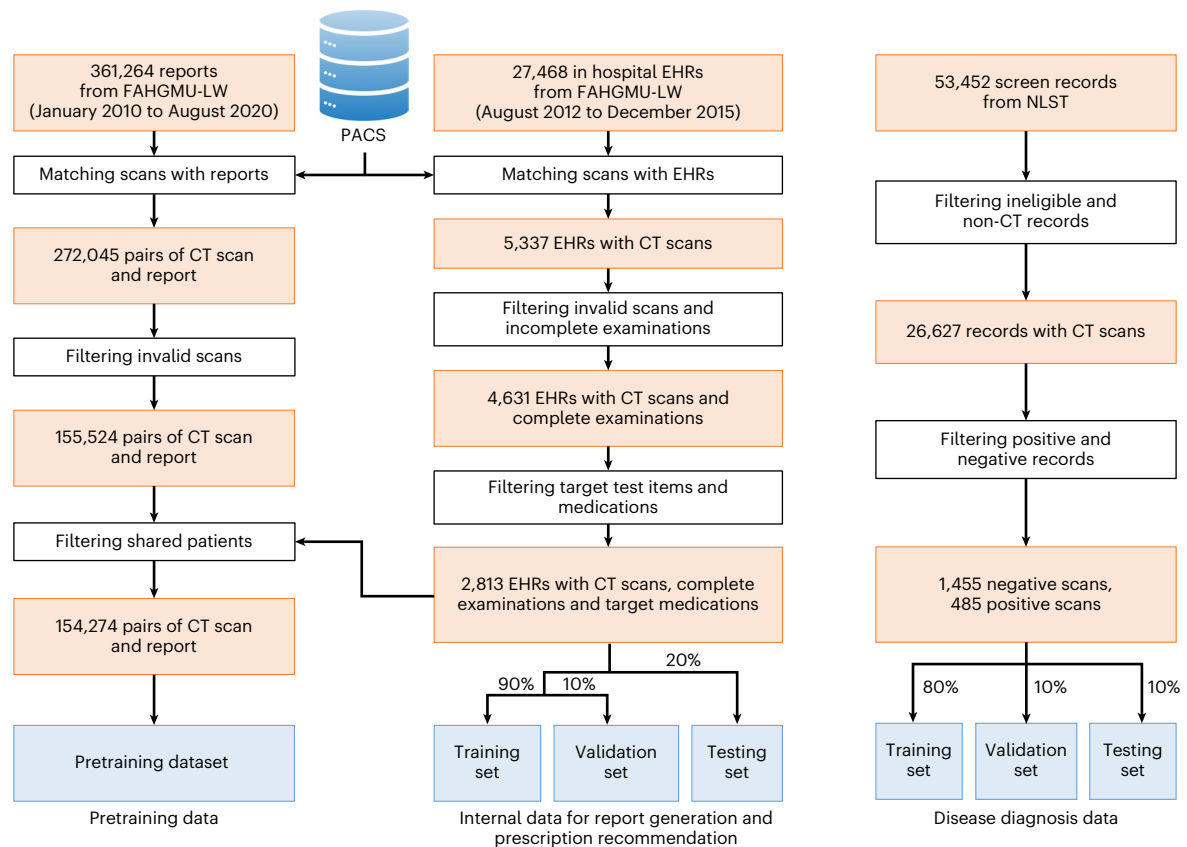


Fig. 7 | The overall data collection process of pretraining and downstream tasks. Left: a total of 361,264 radiology reports (January 2010 to August 2020) were collected from FAHGMU-LW. After matching and post-filtering, a dataset with 154,274 CT scans and paired reports was collected for pretraining. Middle: for the tasks of report creation and prescription recommendation, 27,468 electronic health records (EHRs) (August 2012 to December 2015) were collected from FANGMU-LW. After matching and filtering, a dataset with 2,813 EHRs

containing complete CT scans, radiology reports, demographic information, laboratory tests and target medications was acquired, which was divided into the training, validation and retrospective test sets. Right: a total of 53,452 screen records from the public database NLST were collected and filtered. Finally, 485 CT scans with a diagnosis of lung cancer and 1,455 negative scans were retained and divided into training, validation and test sets to evaluate the disease diagnosis performance.

these models to optimize clinical workflows and facilitate human–AI collaboration in real-world healthcare settings. Also, recent research⁵³ indicates that large language models have the potential to generate malicious or harmful data, leading to data contamination, which in turn can adversely affect model training and downstream applications. This raises concerns about existing models trained on data sourced from open-access publications, specifically whether they possess effective mechanisms to detect and filter out potentially malicious inputs, thereby ensuring data quality and authenticity and preventing model training from being compromised by malicious data.

In this study, we developed MedMPT, which is capable of effectively integrating and processing multiple modalities of heterogeneous medical data and supporting a wide range of clinically aligned tasks. Due to the difficulties in collecting and labelling medical data, a common solution for developing a medical model is to use transfer learning from the models pretrained on large-scale natural data. However, MedMPT highlights the importance of high-quality domain knowledge for achieving more efficient and effective performance in clinical applications. Overall, our comparative experiments show that pretrained models adapted with medical domain knowledge outperform those based solely on general-domain data. The linear-probed MedMPT, which was fine-tuned with only 1,500 parameters for disease diagnosis and 41 million parameters for prescription recommendation, achieved performance comparable to or better than the natural-baseline and OpenAICLIP models, which involved over 86 million and 200 million parameters, respectively. These findings emphasize the importance of incorporating domain-specific knowledge in dealing with medical

tasks. Comparisons with other biomedical pretrained models further reveal the significance of high-quality, domain-specific knowledge in developing effective foundational models for medicine. While most existing models were pretrained on academic data collected from biomedical publications, MedMPT benefited from high-quality CT scan–report pairs sourced from real-world clinical institutions (Fig. 7), ensuring more precise and clinically aligned learning. The radiology reports used for pretraining contain detailed descriptions by human experts, corresponding with specific visual patterns in CT scans and extensive medical terminology, which is crucial for understanding medical conditions and generating accurate, expert-like clinical interpretations. In the report-generation task, MedMPT produced imaging reports that align with the radiologists professionally, enhancing the efficiency and quality of clinical workflows through effective human–AI collaboration. Conversely, Med-Flamingo and LLaVA-Med often produced results irrelevant or inconsistent with clinical needs. These findings underscore the necessity of using high-quality medical knowledge for pretraining to develop models truly applicable in clinical settings.

MedMPT also stands out from existing vision–language pretrained models in the medical domain due to its innovative multiscale, multimodal and self-supervised approach, tailored specifically for medical data and clinical tasks. Unlike models that merely continue pretraining from a general CLIP model using domain-specific data, MedMPT integrates insights derived from clinical experience. It processes sequential CT slices as a unified entity to extract a more comprehensive understanding of patient information and adopts multiwindow input settings to capture variations across different

states, fostering deeper and more accurate analysis of medical images. Moreover, MedMPT enhances self-supervised learning with specific adaptations to address the inherent challenges of medical imaging data. It performs vision self-supervised learning at multiple scales, both at the slice and scan levels, and conducts vision–language contrastive learning in a global context to improve its capabilities and robustness. The superior performance of MedMPT compared with MedMPT(2D) and ablation models clearly shows the effectiveness of these clinically aligned designs. In addition, the multitask vision–language pretraining approach and modular architecture enable MedMPT to handle both discriminative and generative tasks, meeting a wide range of clinical needs. The comparison against the MedMPT(CLIP) baseline and the state-of-the-art models such as PubMedCLIP and BiomedCLIP highlights its applicability and versatility. The outstanding performance and adaptability of MedMPT across diverse clinical tasks highlight the importance of tailored multimodal methods in effectively handling complex medical data and applications within the medical domain. MedMPT exploits multimodal medical data to provide holistic understandings of patients and facilitate diverse clinical tasks, improving the efficiency and effectiveness of medical practice and bringing it closer to the real sense of a general-purpose foundation model.

Despite the comprehensive evaluation of MedMPT across various clinical tasks in our study, there are still certain limitations and challenges that need to be addressed in future research. Our study primarily focuses on respiratory diseases using commonly used modalities, such as chest CT, radiology reports, laboratory tests and demographic information. The generalizability of MedMPT to other modalities has not yet been explored. Expanding the volume and diversity of demographics and populations in the internal and external test set in future studies will ensure more comprehensive and robust evaluations. We will collect more diverse and balanced data covering a broader range of clinical tasks continually and conduct prospective evaluations in future studies to explore the impact and effectiveness of MedMPT more comprehensively. Finally, it is important to acknowledge that MedMPT should be used as a supportive system for human experts, rather than as a replacement. Solely relying on MedMPT to complete clinical tasks independently could lead to unsafe outcomes. Emphasis should be placed on responsibly using MedMPT to prevent potential misuse. The clinical judgement, experience and specialized knowledge of experts remain critical in understanding the nuances of each patient's unique situation. Healthcare providers should always exercise their professional judgement and carefully consider the recommendations from models. By combining the strengths of human expertise and technological advancements, we can strive for a future in which healthcare providers have access to powerful tools that augment their capabilities and improve healthcare outcomes.

In summary, in this study we systematically assessed MedMPT by benchmarking it against advanced vision–language pretrained models in the medical domain, conducting quantitative evaluations, qualitative analyses, human expert evaluations and ablation studies in diverse, clinically aligned applications. MedMPT showed superior performance and applicability in diverse scenarios, highlighting its potential as a promising approach for developing a foundational model in clinical practice. Human expert evaluations indicated that effective human–AI collaboration can lead to more precise diagnoses, well-informed treatment planning and thereby enhance patient outcomes.

Methods

Data collection

An overview of the complete data collection process is shown in Fig. 7. We retrospectively collected 361,264 electronic radiology reports from FAHGMU-LW between 1 January 2010 and 31 August 2020. Using patient identifiers and inspection times, we retrieved 272,045 matched CT scans from the PACS. All CT scans were stored in a digital imaging and communications in medicine (DICOM) format. After excluding

incomplete, over-short, non-plain or non-axial scans, and retaining only the first visit for patients with multiple records, we obtained 155,524 pairs of plain CT scans and radiology reports. To avoid data leakage, we removed patients overlapping with the downstream datasets, yielding a final pretraining dataset of 154,274 CT–report pairs.

We evaluated the performance of MedMPT on three diverse target tasks, that is, disease diagnosis, report generation and prescription recommendation. For disease diagnosis, we used the NLST database²⁰ for lung cancer screening and the public MosMedData²¹ dataset for COVID-19 diagnosis and severity grading. In the NLST database, eligible participants with consistently negative screening results were selected as negative samples, whereas those with positive initial screening and confirmed invasive follow-up examinations were selected as positive samples for constructing the lung cancer screening dataset. After excluding the samples with overly short or overly long scans, we ultimately obtained 1,455 negative and 485 positive scans, which were randomly divided into the training, validation and test sets, with 1,552, 194 and 194 scans, respectively. MosMedData contains 254 scans without COVID-19 symptoms and 856 COVID-19-positive scans, which were further split into 4 categories: CT1 ($\leq 25\%$ lung involvement, 684 scans), CT2 (25–50%, 125 scans), CT3 (50–75%, 45 scans) and CT4 ($> 75\%$, 2 scans). For COVID-19 diagnosis, binary classification was performed following a previous study⁵⁴ to identify the COVID-19 scans. For severity grading, we excluded the CT4 category due to its limited sample size and conducted a three-fold cross-validation experiment to classify the images into CT0 (healthy), CT1 (mild), CT2 (moderate) and CT3 (severe) according to official guidance.

For reporting and prescribing tasks, we retrospectively collected 27,468 electronic health records from FAHGMU-LW (1 August 2012 to 31 December 2015) and retrieved 5,337 corresponding plain CT scans from the PACS. After excluding irrelevant or incomplete data (missing CT scans, reports, demographics, laboratory tests or medications), we obtained 4,631 eligible cases. We further standardized test items and medications, selecting 98 laboratory test items (Supplementary Table 18) along with patient age and sex as structured input, and 55 commonly used medications (Supplementary Table 19). Cases with less than half of the 98 laboratory tests or undesired medications were excluded, yielding a final downstream dataset of 2,813 samples. This dataset was randomly divided into training (2,023), validation (239) and internal test (551) sets without patient overlap with pretraining data. To evaluate generalizability, we also collected 384 external retrospective records with paired CT scans from The First Affiliated Hospital of Guangzhou Medical University at Yuexiu and applied the same filtering procedures, yielding 178 samples with the necessary multimodal information for external validation.

Architecture overview

MedMPT is a unified multimodal pretrained transformer framework with attention mechanisms^{19,26} to capture relationship across modalities. MedMPT consists of four main components: a vision encoder, a text encoder, a vision decoder and a text decoder (Extended Data Fig. 4a). The vision encoder incorporates a slice encoder followed by a slice fusion module to accommodate the multislice nature of CT scans. The text encoder extracts semantic representations from free-text radiology reports. The vision decoder reconstructs masked CT regions to enhance fine-grained visual representation learning. The text decoder generates radiology reports conditioned on encoded visual features. The network structures of the four primary components are shown in Extended Data Fig. 5.

Multitask pretraining algorithm

To acquire multimodal knowledge from large-scale medical data without human annotation, we propose a multitask pretraining algorithm tailored to the unique characteristics of medical data and clinical tasks by jointly optimizing four complementary objectives: intramodal

contrastive learning to explore the non-trivial global features for medical images, intermodal contrastive learning to extract medical insights from image and text perspectives, masked image reconstruction (MIR) exploiting fine-grained information within images, and cross-modal generation for deepening collaborative enquiry between multimodal medical data. The overall pretraining framework is shown in Extended Data Fig. 4a.

Intramodal contrastive learning. To capture meaningful representations from unlabelled data, we tailor a self-supervised contrastive learning approach (image–image contrastive (IIC)), which exploits the inherent structural and contextual relationships within CT images. IIC learns robust visual features by encouraging different views of the same image (positive pairs) to have similar representations, while pushing apart representations of different images (negative pairs). During pretraining, augmented view pairs are encoded into a shared latent space, where the objective of IIC is to maximize similarity between positive pairs and minimize it between negative pairs.

Considering the multislice nature of CT scans, we design a vision encoder comprising a ViT-based²⁶ slice encoder, followed by a self-attention-based slice fusion encoder to integrate inter- and intraslice relationships within a scan. To capture fine-grained details and global anatomical context simultaneously, MedMPT performs contrastive learning at both the slice and scan levels. Specifically, a CT scan volume $X_v = \{s_0, s_1, \dots, s_{N-1}\}$ consists of multiple slices, where s_i denotes the i th slice and N is the total number of slices. Each slice s_i is randomly augmented twice to produce query slices $\{q_i\}$ and positive keys $\{k_{i+}\}$, which are then patchified as input tokens. To reduce redundancy and improve computational efficiency, 20% of patches in $\{q_i\}$ are randomly masked and removed from the input sequences. The visible patches of queries $\{q_i\}$ and all patches of $\{k_{i+}\}$ are projected into embeddings and fed into the network. A query vision encoder E_v and a momentum encoder E_{vm} are used to extract visual representation of a visit. The parameters of E_v are trainable and the parameters of E_{vm} are momentum updated by a moving average of E_v as:

$$\theta_{vm} \leftarrow m\theta_{vm} + (1 - m)\theta_v \quad (1)$$

where θ_v and θ_{vm} denote the parameters of E_v and E_{vm} , and $m \in [0, 1]$ is a momentum coefficient, which is set as 0.996 following previous practice⁵⁵. The momentum encoder makes model training more stable and helps create more reliable outputs. We extract slice-level representations $\{z_i^q\}$ and $\{z_{i+}^k\}$ from the slice encoders and scan-level representations Z^q and Z^k from the output of the slice fusion encoders. The objective of IIC is to maximize the similarity of the queries to the positive keys in both slice and scan levels, that is, z_i^q to z_{i+}^k , and Z^q to Z^k , while dissimilating them to the unmatched keys. Instead of selecting negative keys from large mini-batch⁵⁶, we follow the practice of previous studies^{55,57}, maintaining two feature queues Q_{slice} and Q_{scan} of size 65,536 and 2,048, respectively, to provide negative keys efficiently. We compute slice- and scan-level contrastive loss using InfoNCE Loss⁵⁸ to learn vision embeddings for CT scans:

$$L_{\text{c_slice}} = -\sum_i^N \log \frac{\exp(z_i^q \cdot z_{i+}^k / \tau_{\text{IIC}})}{\sum_j^{K_{\text{slice}}+N} \exp(z_i^q \cdot z_j^k / \tau_{\text{IIC}})} \quad (2)$$

$$L_{\text{c_scan}} = -\log \frac{\exp(Z^q \cdot Z^k / \tau_{\text{IIC}})}{\sum_j^{K_{\text{scan}}} \exp(Z^q \cdot z_j^k / \tau_{\text{IIC}})} \quad (3)$$

$$L_{\text{IIC}} = L_{\text{c_slice}} + L_{\text{c_scan}} \quad (4)$$

where $L_{\text{c_slice}}$ and $L_{\text{c_scan}}$ are the slice-level and scan-level contrastive losses and L_{IIC} represents the overall loss of IIC. z_i^k is sampled from both the N positive keys within the current scan and the K_{slice} negative keys in Q_{slice} , while z_j^k comes from the K_{scan} negative keys in Q_{scan} only.

In our study, N is set as 32 and τ_{IIC} is a learnable temperature parameter initialized to 0.07. Both queues are updated progressively, where the positive keys $\{z_{i+}^k\}$ and Z^k will be enqueued to the corresponding queues and the oldest features will be removed.

Intermodal contrastive learning. Radiology reports provide detailed, semantic information about the CT scan, offering fine-grained and contextual interpretations for understanding CT images. To extract these medical insights, we adopt cross-modal image–text contrastive (ITC) learning²², which aligns CT images with their corresponding reports in a shared latent space by bringing together the embeddings of paired images and reports while pushing apart mismatched images and reports. While most existing vision–language models align reports with 2D slices individually, MedMPT adopts ITC to align the report with the aggregated embeddings of entire scans. This global approach captures the holistic clinical context, improving reliability and applicability in real-world scenarios.

Specifically, each radiology report $X_t = \{t_0, t_1, \dots, t_{M-1}\}$ with M tokens is tokenized and encoded by the text encoder E_t , and its global representation Z^r is represented as the average of output token embeddings. Along with the corresponding CT scan representation Z^q , ITC pulls paired Z^r and Z^q closer while pushing apart unpaired instances within the mini-batch. The objective of ITC is to enhance the cross-modal alignment for knowledge transfer. The bidirectional contrastive loss is presented as:

$$L_{\text{ITC}} = -\frac{1}{2} \left[\log \frac{\exp(Z^q \cdot Z^r / \tau_{\text{ITC}})}{\sum_j^S \exp(Z^q \cdot z_j^r / \tau_{\text{ITC}})} + \log \frac{\exp(Z^r \cdot Z^q / \tau_{\text{ITC}})}{\sum_i^S \exp(z_i^r \cdot Z^q / \tau_{\text{ITC}})} \right] \quad (5)$$

where S is the mini-batch size, z_j^q and z_i^r denote the features of query CT and corresponding report of the j th sample and τ_{ITC} is a learnable temperature parameter initialized as 0.07. ITC enriches semantic understanding of CT images by aligning modalities, fostering deeper comprehension of multimodal data and promoting generalizability in clinical tasks.

MIR. Considering that critical anomalies in medical images can be very subtle and localized, often appearing within complex anatomical structures, we introduce MIR to the CT slice as a supplement to intramodal contrastive learning. MIR requires predicting missing parts of an image from the remaining visible parts, encouraging local attention to promote the representation of fine-grained features.

As previously mentioned, given a set of query slices $\{q_i\}_{i=0, \dots, N-1}$, the visible patches of the randomly masked slices are fed into a slice encoder resulting in the slice representation $\{z_i^q\}_{i=0, \dots, N-1}$. Then, the encoded representations of the visible patches along with the predefined mask token embedding z^{mask} are concatenated and fed into a lightweight decoder D_v . The target of MIR is to reconstruct the masked patches in query slice q_i from the encoded representation z_i^q , which requires the model to focus on the local details and facilitates the extraction of informative representations. Following a previous study⁵⁹, we computed the reconstructed loss on the masked regions to optimize the MIR as:

$$L_{\text{MIR}} = \frac{1}{N} \sum_{i=0}^{N-1} \|q_i - D_v(z_i^q, z^{\text{mask}})\|_{\text{mask}} \quad (6)$$

Compared with the objective of IIC, which focuses on the global alignment of image views, MIR facilitates learning fine-grained features. Incorporating IIC and MIR, we enhance the ability to capture both global and local features, resulting in a more comprehensive representation of CT images.

Cross-modal generation. To further enhance cross-modal understanding and broaden the adaptability of MedMPT, we incorporate

the cross-modal report generation task⁴⁹ into MedMPT. The aim of report generation is to generate a textual report that provides correct and complete descriptions of the input CT. The generated report should accurately capture all relevant details, findings and observations present in the image, which enforces the model to extract a comprehensive understanding of the image and enables it to support the text generation tasks.

In cross-modal generation, we use the slice embeddings output from the vision encoder E_v , which contains both the inter- and intraslice information within a CT scan as the slice representations denoted as $\{z_i^q\}_{i=0, \dots, N-1}$. The informative representations of each slice within a study are then fed into a transformer decoder-based text generator D_t as a query and key for cross-attention computation to conduct sequence-to-sequence learning. We use the cross-entropy loss, which maximizes the likelihood of the target text in an autoregressive manner to optimize the model. Therefore, given the slice representations $\{z_i^q\}$ and corresponding report $X_t = \{t_0, t_1, \dots, t_{M-1}\}$, the objective is to maximize the conditional log-likelihood probability of the predicted report formulated as:

$$L_{\text{CMG}} = - \sum_{i=1}^{M-1} \log P(t_i | t_0, \dots, t_{i-1}, z_0^q, \dots, z_{N-1}^q) \quad (7)$$

where CMG refers to cross-modal generation, N is the number of slices within a scan and M is the length of the report tokens. Maximizing the likelihood of the target text encourages the model to generate captions that accurately describe the visual content of the images, aligning the textual output with the semantic information present in the images.

Fine-tuning the algorithm for downstream tasks

In this section, we present the fine-tuning algorithms for the downstream tasks. We first describe the task-specific methods (Extended Data Fig. 4b) for image diagnosis, report generation, and prescription recommendation, and then introduce a multitask, joint fine-tuning strategy.

CT-based disease diagnosis. To adapt MedMPT to image-based diagnosis, we use the pretrained vision encoder with a task-specific classification head to map visual representations to disease categories. We perform both end-to-end fine-tuning and linear probing (freezing the encoder and training only a linear head) to evaluate the performance. End-to-end fine-tuning enables the pretrained model to integrate general knowledge from pretraining with task-specific patterns to improve its performance on target tasks, while linear probing assesses the effectiveness and generalizability of the learned representations.

Report generation. In reporting tasks, we apply the pretrained vision encoder and the text decoder to facilitate the generation of accurate and informative reports. Similarly, we first conduct end-to-end fine-tuning using the training set of the downstream task to adapt the pretrained parameters to the target domain. As report creation perfectly aligns with the objective of the cross-modal generation task in the pretraining phase, we can evaluate the performance of MedMPT in a zero-shot manner, without any fine-tuning or adaptation on the labelled data specific to the target domain, showing the generalizability and data efficiency of MedMPT.

Prescription recommendation. As one of the most complex tasks in clinical practice, prescribing requires a comprehensive analysis of the patient's conditions from multiple perspectives, supported by sufficient medical knowledge to provide appropriate prescriptions. Therefore, we incorporate the pretrained vision and text encoder along with a structured data encoder to process CT images, reports

and structured data (demographics and laboratory tests). These features are fused via a multimodal encoder to acquire a comprehensive representation of each case. The structured encoder and multimodal encoder are built using attention mechanisms¹⁹ to capture the relationships between individual indices and modalities. We initialized the additional modules from scratch and conducted both end-to-end and partial (freezing pretrained parameters and updating the remaining parameters) fine-tuning to evaluate the adaptability of MedMPT.

Considering the significance of medication correlations, we constructed a medication graph to capture prior knowledge from existing prescriptions. Specifically, we retrieved 27,468 in-hospital electronic health records from PACS (Fig. 7) and selected 17,060 prescriptions (non-overlapping with the test set) to compute the co-occurrence matrix (Supplementary Fig. 5), where each entry reflects the frequency of two medications appearing together. We model these relationships as a medication graph in which each node represents a medication and each edge indicates the co-occurrence probabilities of two connected medications. Edges with low co-occurrence probabilities were pruned to reduce noise. On the basis of this graph, we built a multi-scale dynamic graph attention network⁶⁰ with four stacked attention layers, alternately using a fully connected graph and a sparse graph retaining only high-probability edges to balance detailed interactions and prior patterns (Fig. 6b). The encoded medication features were concatenated with patient features and fed into a linear classifier to generate personalized prescription recommendations.

Multitask joint fine-tuning. While task-specific fine-tuning allows the model to learn and optimize parameters specifically for each individual task, a unified fine-tuning model would be beneficial in improving the clinical utility of the AI model and enhancing efficiency in clinical practice. Therefore, we further explored multi-task joint learning to adapt MedMPT to diverse clinical tasks simultaneously. We construct a unified network derived from the pretrained MedMPT, comprising a unified vision encoder, a text decoder for report creation and a text encoder for prescription recommendation. Remaining task-specific modules were initialized from scratch and fine-tuned using a combination of training sets from lung cancer screening, report creation and prescription recommendation tasks.

Comparisons and baselines

We compared MedMPT with state-of-the-art vision–language pretrained models in the medical domain. OpenAI CLIP²² uses 400 million web-sourced (image and text) pairs for vision–language contrastive learning and serves as a general multimodal baseline. PubMedCLIP²³ fine-tunes CLIP using approximately 81,000 medical image–text pairs from the ROCO dataset⁶¹. BiomedCLIP¹² performs vision–language contrastive pretraining on the PMC-15M dataset, which comprises 15 million biomedical image–text pairs from 4.4 million scientific articles. Med-Flamingo²⁴ is trained to generate open-ended answers for medical visual questions with an interleaved dataset from a set of medical textbooks and a biomedical image–caption dataset collected from PubMedCentral's OpenAccess subset. LLaVA-Med²⁵ adapts LLaVA⁶² to the biomedical domain, using multimodal instruction-following data from PMC-15M and GPT-4 to create a multimodal conversational assistant for biomedical applications. We also include task-specific baseline models using ViT pretrained on ImageNet²⁷ or pretrained BERT³¹ to serve as natural baselines. We also designed two MedMPT variants to evaluate key designs. MedMPT(2D) retains multitask pretraining but removes the slice fusion module, performing contrastive learning on individual slices with its report instead. MedMPT(CLIP) follows a CLIP-like approach, conducting slice-level contrastive pretraining only, without a multitask objective. Both variants use the same pretraining dataset as MedMPT. Detailed descriptions and analysis of the comparative models are provided in Supplementary Methods.

Implementation details

To facilitate effective learning of clinically relevant features, we applied a series of preprocessing steps. CT scans were resampled, uniformly sampled to 32 slices per case. Each slice was resized to 512×512 and processed using 3 lung and mediastinum windows, that is, (window level -400, window width 1,500), (-500, 2,000) and (60, 400). The 3 windowed images were concatenated as a $32 \times 512 \times 512 \times 3$ input to better highlight pathological information. Standard data augmentations were used during training to improve robustness. Reports were cleaned via regular expressions and laboratory tests were normalized by reference ranges. For explainability, we applied the attention rollout technique^{26,63,64} to display the model attention distribution. In diagnosis and medication tasks, pixel- and slice-level self-attention was visualized, while in report generation, self- and cross-attention were combined to generate word-specific attention maps. We use AdamW optimizer⁶⁵ and cosine annealing scheduler for model training, with an initial learning rate of 1×10^{-6} and a weight decay of 0.05 during pretraining. MedMPT was pretrained for 32 epochs on 8 Tesla V100 GPUs. Downstream models were trained for 50 epochs using an early stopping strategy. All experiments are conducted using PyTorch⁶⁶ and the evaluation metrics are computed using scikit-learn⁶⁷. Further implementation details about data preprocess, attention visualization and training strategies are provided in Supplementary Methods.

Experimental set-up for human–AI collaboration

In this study, we conducted human–AI collaboration experiments to assess the impact of MedMPT on the routine work of radiologists. We randomly sampled 52 cases from internal test sets. In the first stage, we evaluated whether MedMPT-assisted reports could reduce reporting time and improve efficiency. To minimize bias and ensure comparability, test data were randomly divided into two groups and radiologists were paired by years of experience. Within each pair, one radiologist used AI-generated drafts as a reference for group A and a standard template for group B, while the other followed the opposite arrangement. All reports were completed by typing. In the second stage, an additional radiologist reviewed and rated the reports to assess whether MedMPT improved report quality or introduced potential drawbacks.

Evaluation protocols

Quantitative metrics. We used accuracy and AUROC to evaluate diagnosis performance. Accuracy measures the proportion of correctly classified instances out of the total instances. AUROC refers to the area under the ROC curve, which plots the true positive rate against the false positive rate at various threshold settings. It is calculated using the trapezoidal rule on the ROC curve.

For report generation, we used caption metrics to evaluate generated reports. BLEU-n scores measure n-grams precision:

$$\text{BLEU-n} = \exp\left(\sum_{i=1}^n \frac{1}{n} \log p_i\right) \times \text{BP} \quad (8)$$

where p_i is the precision of i -grams and BP is the brevity penalty. METEOR evaluates the alignment by considering matches, stemming, synonyms and word order:

$$\text{METEOR} = F_{\text{mean}} \times (1 - \text{penalty}) \quad (9)$$

where F_{mean} is the harmonic mean of precision and recall, and penalty is a function of the fragmentation of matched unigrams. ROUGE-L measures the longest common subsequence:

$$\text{ROUGE-L} = \frac{(1+\beta^2) \cdot R \cdot P}{R+\beta^2 \cdot P} \quad (10)$$

where R is recall, P is precision and β is typically set to 1.

For prescription recommendation, precision calculates the proportion of true positive (TP) out of all positive predictions made by the model as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (11)$$

where FP refers to false positive. Recall measures the proportion of true positive out of all actual positives as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (12)$$

where FN refers to false negative. F1 score provides a single metric that balances both precision and recall:

$$\text{F1 score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (13)$$

Evaluation criteria for reports quality assessment. In the report-generation task, we invited a senior radiologist to review and score the reports generated with and without MedMPT assistance. Quality is evaluated across 3 aspects, readability, accuracy and completeness, using a 1 to 10 scoring system, where higher scores represents better performance. Readability assessed grammar and logic, accuracy evaluated consistency with CT findings and completeness measured inclusion of all relevant observations. Detailed criteria are provided in the Supplementary Methods.

Evaluation criteria for prescription quality assessment. In the prescription recommendation task, we engaged 4 junior clinicians to assess recommended prescriptions on a 6-point scale as follows: 5 for complete coverage of all necessary medications without any DDIs, 4 for covering at least 50% of required medications without DDIs, 3 for complete coverage of all required medications with DDIs, 2 for covering less than 50% of required medications without DDIs, 1 for covering at least 50% of required medications with DDIs, and 0 for covering less than 50% of medications with DDIs. The evaluated results were confirmed by two senior clinicians.

Statistical analysis

We used the SciPy package in Python for statistical analysis. Specifically, we computed the 95% CIs using bootstrap resampling 1,000 times from the test set. To assess statistical significance, we used the two-sided Wilcoxon rank-sum test to calculate P values. A P value < 0.05 was considered statistically significant.

Ethics statement

The study was approved by the Ethics Committee of the National Center for Respiratory Medicine/The First Affiliated Hospital of Guangzhou Medical University (approval number 2022070). The need for informed consent was waived due to the retrospective nature of the study.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data from NLST are available at <https://cdas.cancer.gov/datasets/nlst/>. Data from MosMedData are available at <https://github.com/michaelwfr/MosMedData-Chest-CT-Scans-with-COVID-19-Related-Findings/>. The remaining data collected from The First Affiliated Hospital of Guangzhou Medical University are not publicly available due to privacy requirements. A de-identified validation set can be made available for research purposes upon reasonable request to the corresponding authors.

Code availability

The code for pretraining and fine-tuning of MedMPT, along with the pretrained model weights can be found at GitHub at <https://github.com/maliangdi/MedMPT> (ref. 68).

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Author contributions

F.X., J.H., Y.G. and H.L. are the co-corresponding authors. L.M., H.L., Y.H., J.H., Y.G. and F.X. contributed to the conception and design of the research. F.X., J.H. and Y.G. coordinated and organized the research team to complete this work. H.L. and W.W. contributed to the raw data acquisition. L.M., Y.H. and Z.Y. contributed to data organization and verification. L.M., Y.G. and F.X. contributed to the methodology, technical implementation and results analysis. H.L., Y. Liu and J.H. organized the clinical team for the validation experiment. H.L., W.W., Y. Li, W.L., R.W., Y. Lizhu, Y. Liu and J.H. contributed to the clinical validation experiment and results evaluation. L.M., Y.G. and F.X. contributed to the original draft preparation and revising of the paper. L.M., H.L., Y.G., J.H. and F.X. discussed the results and commented on the paper.

Competing interests

Tsinghua University has filed for patent protection for F.X., Y.G. and L.M. for the work related to the multimodal pretraining method in chest CT. The other authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41551-025-01544-z>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41551-025-01544-z>.

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Example 1

Ground truth: Both lungs show uneven translucency and thickened pulmonary markings. Multiple ground-glass blurred shadows and reticular patterns are seen beneath the pleura in both lungs, with some subpleural lines formed. No stenosis is observed in the trachea or bronchi. No enlarged lymph nodes are seen in the lung hila or mediastinum. The cardiac silhouette is not enlarged, and there is no pleural thickening. The chest wall shows no deformities, and no soft tissue masses are observed. No obvious bone destruction is noted in the ribs and sternum within the scanned range.

MedMPT: The texture of the left lung is thickened and disordered; Multiple small patchy blurred shadows, increased ground-glass density shadows, and fibrous stripe shadows are seen in both lungs. Most lesions show grid-like changes, mainly in the subpleural area and both lower lungs. The interlobular septa of both lungs are thickened, and a subpleural line is visible. The lesions vary in size. The hila of both lungs are not large. No enlarged lymph nodes are seen in the mediastinum. The shadow of the heart is not large. No clear destruction of bone is observed.

Med-Flamingo: The transparency of both lungs decreases, and the lung markings become thicker and more disorderly; Multiple small patchy blurry shadows, increased ground glass density shadows, and fibrous cord shadows were observed in both lungs. Most lesions showed grid like changes, mainly in the subpleural area and in the lower lungs, with thickened interlobular septa between the two lungs. The two lung hili are not large. No enlarged lymph nodes were found in the mediastinum. The heart is full of shadows. The bone shown is not clearly damaged. Pulmonary angiography reconstruction showed that the main pulmonary artery, left and right pulmonary artery trunks, various lobe segments, and branches below the observed segments were unobstructed, with no stenosis, dilation, or filling defects, and no abnormal vascular clusters. The scanning showed a general decrease in liver density, with a CT value of 43/50Hu for the liver/spleen at the same level. Multiple cystic low-density shadows of varying sizes were observed in the liver, with clear boundaries. The larger ones had a diameter of about 2.3cm. The gallbladder wall is slightly thickened, with a CT value of about 31/50Hu. No abnormalities were found in the pancreas. The left adrenal gland is slightly thickened, with a CT value of about 34/50Hu. The left kidney is slightly larger than the right kidney, and the CT value is about 33/50Hu.

LLaVa-Med: The volume of the right kidney has decreased, and the right renal artery, renal vein, and branches below the renal hilum are significantly narrowed, with the main trunk being about 1.3x narrower than before, and no stenosis was found in the remaining branch of the upper left kidney. The CT value of the right kidney is reduced, and the cortical bone of the right kidney is not thickened. A type of nodular soft tissue mass is seen in the right renal hilum, with a size of approximately 1.0 * 0.8cm. The CT value is approximately 30Hu, and there is no significant enhancement in the enhanced scan. The mass is enveloped by a thin, smooth capsule. A small amount of left pleural effusion accumulates. Multiple thin-walled, small, and patchy transparent areas were observed in the liver, with a maximum diameter of approximately 0.5cm. A type of circular low-density shadow with a diameter of approximately 1.2cm is seen in the S8 segment of the liver, with clear boundaries. A type of circular low-density shadow with a diameter of approximately 0.4cm is seen in the S7 segment of the liver, with clear boundaries. The CT value of the liver is reduced.

Example 2

Ground truth: A tiny nodule is seen in the apical segment of the right upper lung, with clear borders. No obvious abnormal density is seen in the remaining lung fields. No enlarged lymph nodes are observed in the lung hila or mediastinum. The cardiac silhouette is not enlarged. No pleural thickening is seen. No pleural effusion is present in either hemithorax. No obvious bone destruction is noted in the ribs or vertebrae.

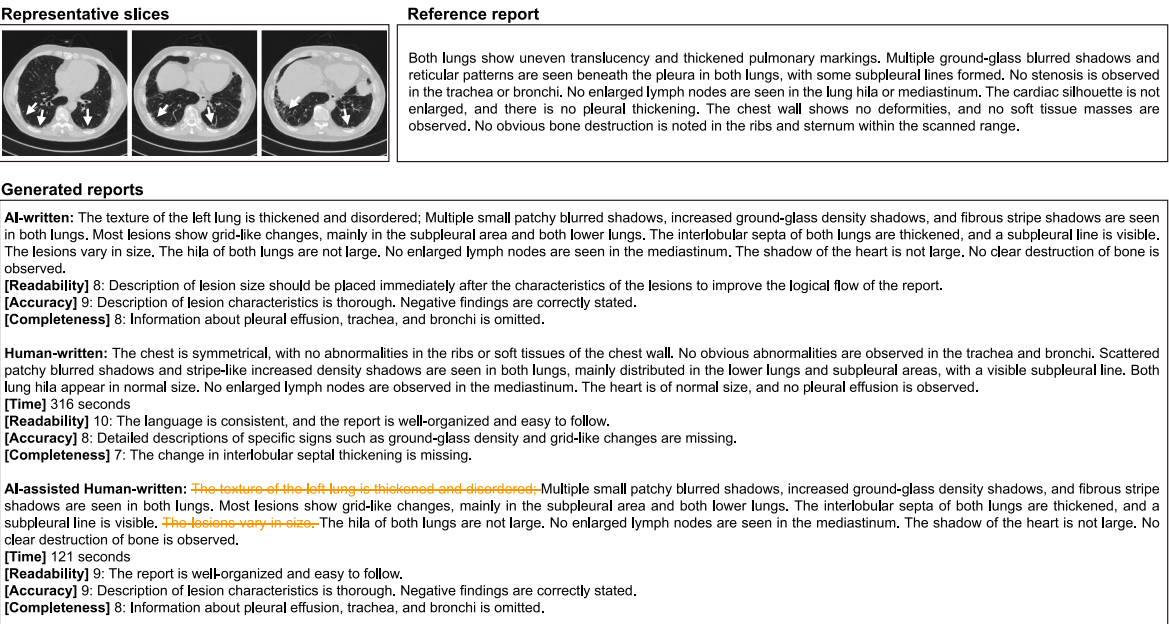
MedMPT: A ground-glass nodule is seen in the posterior segment of the right upper lung apex with clear borders. No obvious lesions are observed in the remaining lung parenchyma. There are no signs of stenosis or dilation of the trachea and bronchi, no enlargement of the hila on either side, and no enlarged lymph nodes in the mediastinum. There is no enlargement of the heart shadow, and no pleural thickening is noted. The chest is not deformed, with no soft tissue masses on the chest wall, and no obvious abnormalities are found in the ribs and sternum within the scanning range.

Med-Flamingo: The lung fields on both sides are clear, with natural lung markings and no clear lesions in the lung parenchyma. There is no stenosis or dilation in the trachea and bronchi, no enlargement in the pulmonary hilum on both sides, and no enlarged lymph nodes in the mediastinum. The heart shadow did not increase, and there was no thickening of the pleura. There is no deformity in the chest, no soft tissue mass is found on the chest wall, and no clear abnormalities are found in the scanning range of the ribs and sternum. A drainage tube shadow is seen in the right chest cavity. The scan shows a small amount of fluid accumulation in the right chest cavity. There is no deformity in the chest, no soft tissue mass is found on the chest wall, and no clear abnormalities are found in the scanning range of the ribs and sternum. The scanning shows a drainage tube shadow in the right chest cavity. No clear abnormalities were found in the remaining scanning areas. A small amount of liquid density shadow is seen in the right chest cavity. The trachea and bronchi are unobstructed, with no obvious narrowing or dilation observed. No clear enlarged lymph nodes were found in the mediastinum. The heart shadow did not increase, and there was no thickening of the pleura. No abnormalities were found in the soft tissue and bone of the chest wall.

LLaVa-Med: The transparency of both lungs increases, roughly the same as before. The lung texture of both lung fields is uniform, and the lung markings are slightly increased, with small patchy blurry shadows in the lungs and thickening of the interlobular septa in some areas. The right diaphragmatic surface is slightly elevated and the right lobe of the liver is enlarged and nodular. Multiple high-density shadows can be seen in the right lobe of the liver, with the largest being approximately 1.3cm x 0.9cm. The CT value on plain scan is about 45/20HU. The gallbladder is locally thickened and filled with a small amount of bile. There is no stenosis or dilation in the bile duct. The heart shadow did not show any abnormalities.

Extended Data Fig. 1 | Qualitative comparison of AI-generated radiology reports. Two cases illustrating the ground-truth (human-written) radiology reports and the corresponding reports generated by MedMPT, Med-Flamingo,

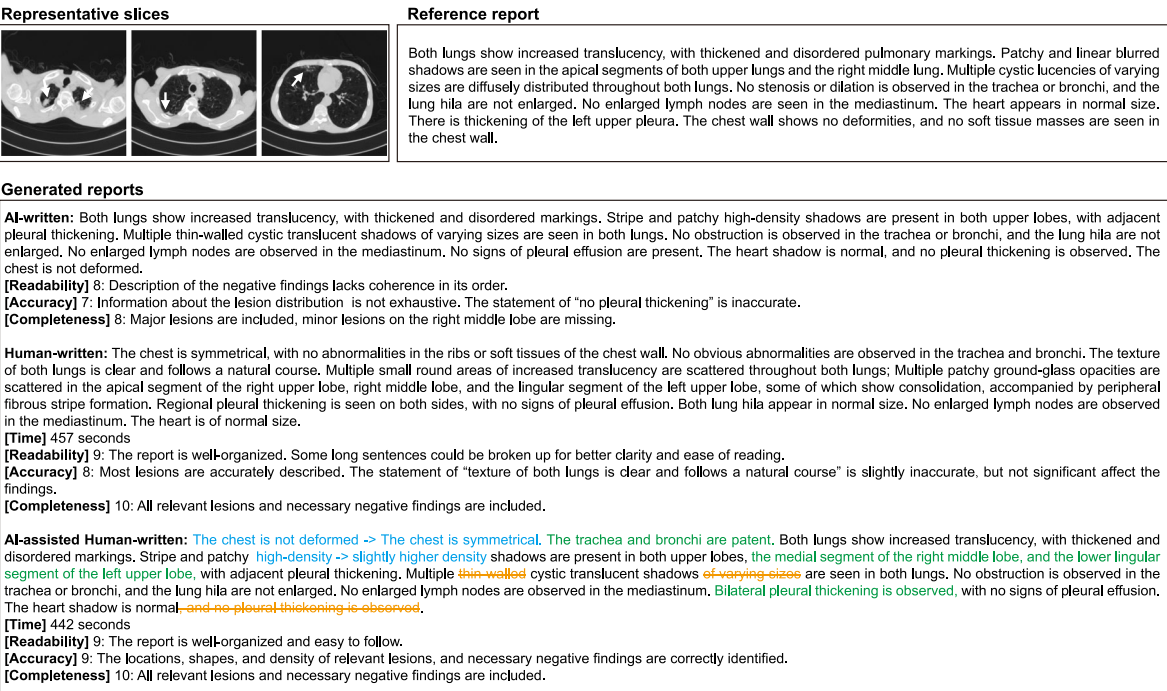
and LLaVa-Med. MedMPT generated accurate and clinically appropriate descriptions, whereas the comparison models produced reports with irrelevant or inconsistent content.



Modifications made by radiologists to the AI's report during the writing process: Deletion Addition Revision

Extended Data Fig. 2 | Case study of human-AI collaboration in report generation. The figure presents the reference report, representative CT slice(s) highlighting key findings, and reports generated by MedMPT (AI-written), a radiologist alone (human-written), and a radiologist assisted by AI (AI-assisted

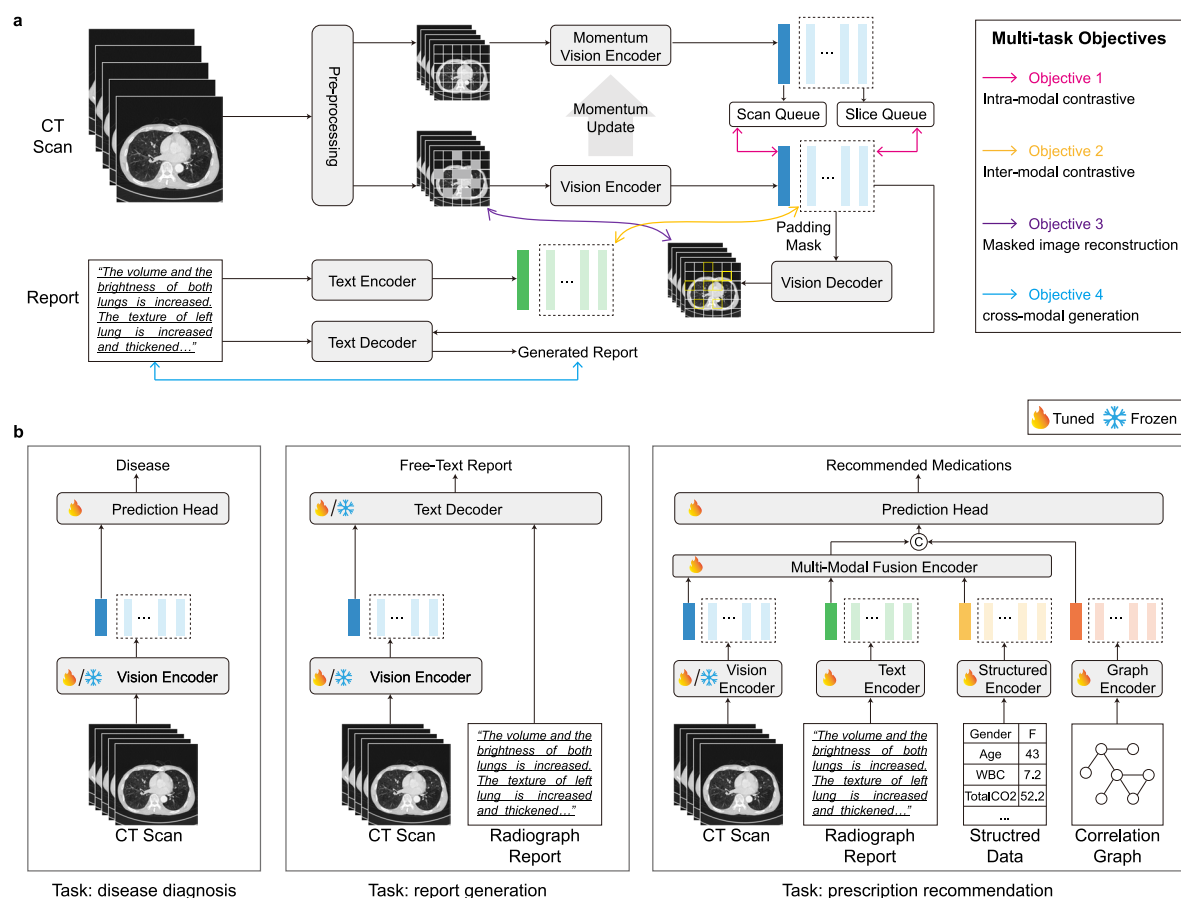
human-written). Radiologist revisions are marked with colour-coded annotations. The AI-assisted report combined the detailed findings from MedMPT with the radiologist's refinements, achieving high quality and fluency while substantially reducing reporting time.



Modifications made by radiologists to the AI's report during the writing process: Deletion Addition Revision

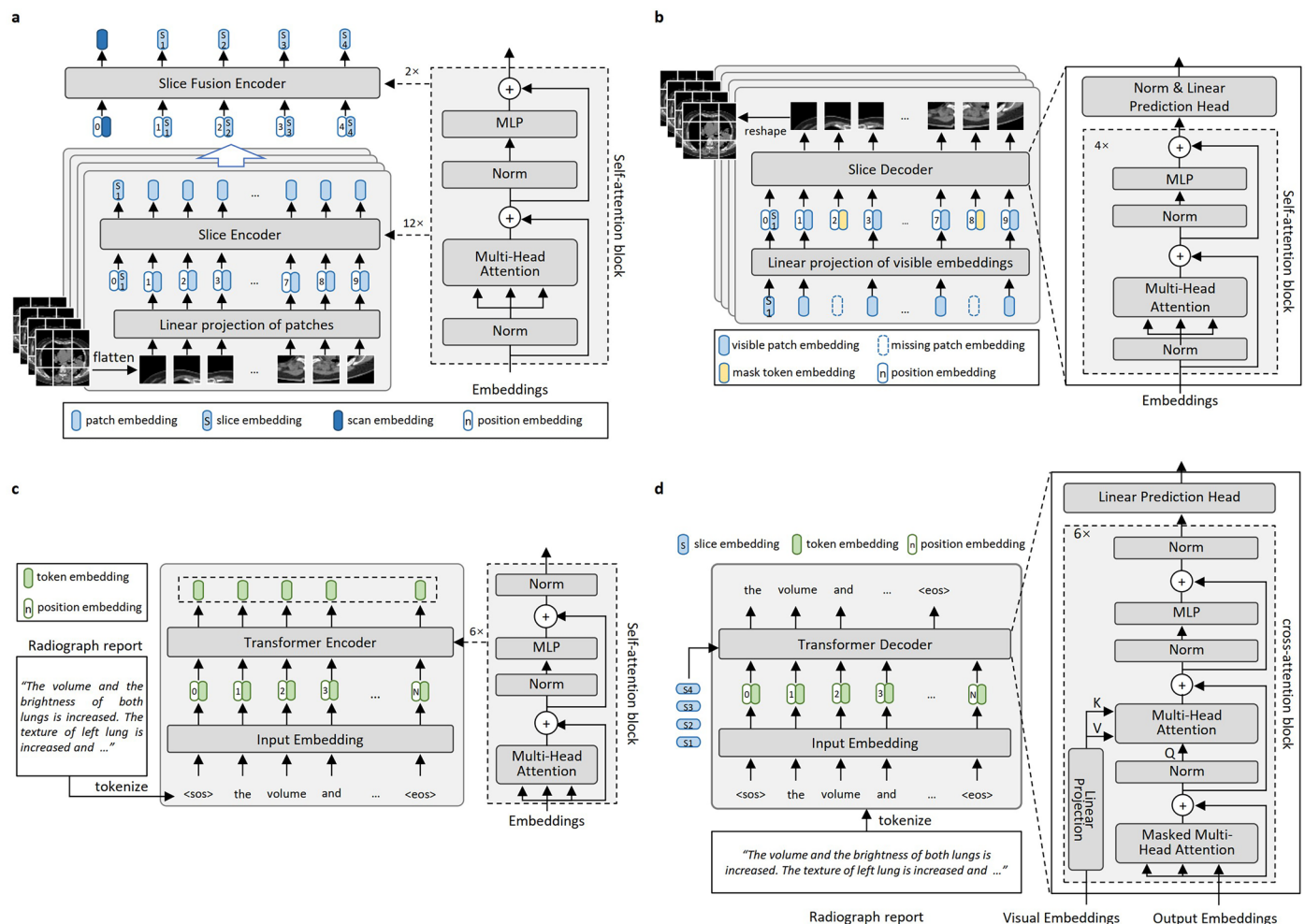
Extended Data Fig. 3 | Case study of human-AI collaboration in report generation. The figure presents the reference report, representative CT slice(s) highlighting key findings, and reports generated by MedMPT (AI-written), a radiologist alone (human-written), and a radiologist assisted by AI (AI-assisted

human-written). Radiologist revisions are marked with colour-coded annotations. In this case, the radiologist revised the AI-generated report to align with their personal writing style, even though it was already clinically accurate, resulting in only a modest improvement in reporting time.



Extended Data Fig. 4 | Overview of MedMPT pretraining framework and finetuning strategies. a, The pretraining framework of MedMPT, where paired CT scans and reports were used to train the vision encoder, vision decoder, text encoder, and text decoder in a multi-task pattern to extract the multi-scale

representations of multi-modal medical data. **b**, The transferring framework of MedMPT to downstream tasks. The pretrained modules, along with additional modules, were employed to facilitate multi-modal downstream tasks. The pretrained parameters can be fine-tuned or frozen for evaluation.



Extended Data Fig. 5 | Overview of the MedMPT Transformer Architecture.

a, Vision encoder, which consists of a slice encoder encoding a series of image patches within a slice as a slice embedding and a sequence of patch embeddings, and a slice fusion encoder encoding the output slice embeddings from the slice encoder within a scan into a global scan embedding and a sequence of slice

embeddings. **b**, Vision decoder, decoding the output patch embeddings of slice encoder (added with mask token embeddings if masks were applied on the input) to a sequence of image patches. **c**, Text encoder, encoding a radiology report as a sequence of token embeddings. **d**, Text decoder, predicting the following texts based on the prefix contents and the slice embeddings from the vision encoder.

Extended Data Table 1 | Data characteristics of pretraining and downstream datasets

	Pretraining set (n=154,274)	Internal set (n=2,813)	External set (n=178)
Age, years			
<40	32,191 (20.87%)	284 (10.09%)	14 (7.87%)
40-59	61,202 (39.67%)	987 (35.09%)	58 (32.58%)
60-79	54,797 (35.52%)	1,350 (47.99%)	95 (53.37%)
>79	6,084 (3.94%)	192 (6.83%)	11 (6.18%)
Gender			
Male (%)	85,150 (55.19%)	1,840 (65.41%)	118 (66.29%)
Female (%)	69,124 (44.81%)	973 (34.59%)	60 (33.71%)

Extended Data Table 2 | Comparison of task-specific and multi-task joint fine-tuning methods across three clinical tasks

Fine-tuning Strategy	Cancer Screening (AUROC)	Report Generation (BLEU4)		Prescription Recommendation (F1 Score)	
		Internal set	External set	Internal set	External set
Task-Specific fine-tuning	0.9269 (0.9260, 0.9288)	0.2123 (0.2004, 0.2231)	0.2081 (0.1901, 0.2296)	0.4576 (0.4565, 0.4580)	0.4262 (0.4249, 0.4276)
Joint fine-tuning	0.9275 (0.9263, 0.9288)	0.1943 (0.1837, 0.2055)	0.1959 (0.1783, 0.2142)	0.4488 (0.4484, 0.4498)	0.4370 (0.4361, 0.4388)
p-value	No Significance	p < 0.0001	p < 0.0001	p < 0.0001	p < 0.0001

The multi-task model demonstrated better generalizability compared to task-specific models. Values are presented as mean with 95% CIs (bootstrapped with 1,000 resamples). The p-values are calculated with two-sided Wilcoxon rank sum test.

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Software and code

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Data collection	Public data from NLST is downloaded from (https://cdas.cancer.gov/datasets/nlst/) via NBIA Data Retriever. Data from MosMedData is downloaded manually from (https://github.com/michaelwfr/MosMedData-Chest-CT-Scans-with-COVID-19-Related-Findings/) manually. Additional data sets supporting the findings of this study are collected manually from the PACS system at the First Affiliated Hospital of Guangzhou Medical University.
Data analysis	All data processing, experiments, and result analysis were completed using Python (version 3.6). PyTorch (version 1.10.2), Transformers (version 4.4.2), and timm (version 0.5.4) were used to build the deep learning framework. Measurement was calculated using Scikit-learn (version 0.24.2). Scipy (version 1.5.4) was used for the statistical analysis of the results. Matplotlib (version 3.3.4), Adobe Illustrator (version 1.0), and Cytoscape (version 3.10.0) were used to plot the figures. The source code of MedMPT can be found in https://github.com/maliangdi/MedMPT .

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Data from NLST are available at <https://cdas.cancer.gov/datasets/nlst/>. Data from MosMedData are available at <https://github.com/michaelwfr/MosMedData-Chest-CT-Scans-with-COVID-19-Related-Findings/>. The remaining data collected from the First Affiliated Hospital of Guangzhou Medical University are not publicly available due to privacy requirements. A de-identified validation set can be made available for research purposes upon reasonable request to the corresponding authors.

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Reporting on sex and gender

The data in this study contains no bias on sex and gender. The data sets collected from the First Affiliated Hospital of Guangzhou Medical University (Guangzhou, China) included all the relevant samples during the pre-established period with no other covariant. The gender of each patient was based on self-reporting. Information on gender was collected and involved in the input to the network in the prescribing task. Gender-based analyses were performed in supplementary information.

Reporting on race, ethnicity, or other socially relevant groupings

Information on race, ethnicity, or other socially relevant groupings was not collected.

Population characteristics

Data retrospectively collected between January 1, 2010, to August 31, 2020, from the First Affiliated Hospital of Guangzhou Medical University contained the entire related population admitted to the institutions with suspected respiratory disorders. Patient ages were categorized into four groups: under 40 (21%), 40–59 (40%), 60–79 (36%), and 80 or older (4%). Both male and female patients were included (male: 55%, female: 45%). Clinical diagnosis was not applicable in this study. Detailed patient characteristics can be found in Extended Data Table 1. The characteristics of the participating radiologists were provided in Supplementary Table 8.

Recruitment

No patient recruitment was performed.

Ethics oversight

The study was approved by the Ethics Committee of the National Center for Respiratory Medicine/The First Affiliated Hospital of Guangzhou Medical University.

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Sample size

The development and evaluation data was retrospectively collected from the First Affiliated Hospital of Guangzhou Medical University (January 1, 2010 - August 31, 2020), and public data (NLST and MosMedData for CT-based disease diagnosis). The sample size for pretraining is 154,274. The sample sizes for lung cancer screening and COVID-19 identifying and severity grading are 1,940 and 1,100, respectively. The sample size for the tasks of reporting and prescribing is 2,813. No formal sample size calculation was performed. Formal sample size calculation methods for machine-learning studies are currently not well-established due to the complexity and variability of models and data. Therefore, sample sizes in this study were determined based on data availability and prior similar studies demonstrating sufficient performance with comparable datasets.

Data exclusions

During data collection, we excluded the samples with incomplete, over-short, non-plain, or non-axial section CT scans, and those with incomplete multi-modal information. To ensure data quality and clinical relevance, we excluded samples with insufficient input items and non-target medications. These criteria were predefined based on domain knowledge and data availability, and were applied consistently during data collection. No additional exclusions were made during the experimental phase. Detailed data inclusion and exclusion descriptions can be found in Methods.

Replication	The model development and evaluation were conducted using randomly split training and test datasets. All procedures are reproducible using the provided code and data. Replication was not applicable to the human reader study, which was designed as a single-session evaluation with a fixed set of participants and counterbalanced AI conditions.
Randomization	For model development, all data were randomly split into training and testing sets using stratified sampling approach. For human evaluation and human-AI collaboration experiments, the involving cases were randomly selected from the full test set. Participating radiologists were randomly assigned to two groups to conduct counterbalanced experiment.
Blinding	The involving radiologists were blinded to patient identifiers, but were aware of patient age, sex, and whether AI assistance was provided.

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Methods

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